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## Effects of talker-to-listener distance on tone

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### ABSTRACT

This paper explores the effects of talker-to-listener distance (TLD) on tone and examines its impact on tone perception by both native listeners and second language learners.

Speakers naturally adjust vocal effort to talk to people at different distances, which leads to changes not only in intensity, duration and formant frequencies, but also in fundamental frequency ( $f_0$ ). Fundamental frequency is the primary acoustic cue that differentiates Mandarin lexical tones from one another. This study aims to answer the question of whether changes in  $f_0$  as a function of TLD affect tone perception by native (L1) and second language (L2) listeners. If so, what are the specific changes that have an impact on tone perception?

The production study investigates the acoustic correlates of the effects of TLD on tone, using 7959 monosyllabic Mandarin speech files recorded by three speakers under 11 levels of TLD. The perception study explores the effects of TLD on L1 and L2 tone perception by 2 native listeners and 52 L2 listeners.

The effects of TLD on speech production are systematic, and we present models and analyses with sufficient detail to simulate these effects. Intensity, duration, and initial and maximum  $f_0$  increase along with TLD, while time-normalized toneshapes remain invariant. The results of the perception study show that native listeners' performance is robust under changes in TLD, while L2 listeners' perception of tone interacts with TLD. One significant finding is that tone 3 recognition improves with TLD.

This work investigates the relationship between speech production and perception when acoustic attributes change naturally in response to the demands of speech communication. The results have potential applications in speech synthesis, pronunciation training and second language testing.

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## 1. Introduction

This paper explores the effects of talker-to-listener distance (TLD) on tone and examines how speech production under demanding conditions impacts tone perception by both native listeners and second language learners.

The distance between a talker and a listener has a strong impact on face-to-face speech communication because acoustic power decreases quickly along with distance. In an open field, acoustic power drops by a factor of 4, or 6 dB, for every doubling of the distance following the inverse square law. In anticipation of the rapid decrease in sound level that occurs when distance increases, talkers intuitively adjust their vocal output to account for that distance, and listeners have matching expectations. Adjustment in speech to account for TLD is a listener-centered adaptation whose goal is successful communication. In principle, everything else being equal, TLD-influenced changes in speech should enhance speech perception, given that this is presumably the intention of the speaker. Whether TLD effects actually help speech perception depends on whether the adjustments made in response to TLD can compensate for the demanding or suboptimal communicative environment created by such long distances. A further complication is the possibility that these adjustments could alter acoustic cues that are crucial for the identification of sound categories, and thus listeners may not be able to successfully decode all such changes.

Previous research has identified the adjustment of vocal effort as the primary talker response to TLD, and changes in vocal effort lead to changes in the acoustic attributes of speech (Cheyney, Kalgaonkar, Clements, & Zurek, 2009; Pickett, 1956; Pickett & Pollack, 1958). Studies of TLD effects and vocal effort have been investigated across segmental and suprasegmental levels, including isolated vowels (Eriksson & Traünmüller, 2002; Liénard & Benedetto, 1999), single words, word pairs or short phrases (Brungart &

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Scott, 2001; Cheyne et al., 2009; Traünmüller & Eriksson, 2000), real conversations (Pelegrín-García, Smits, Brunskog, & Jeong, 2011), different ranges of TLD, and different languages including French, Swedish and English (Cheyne et al., 2009; Liénard & Benedetto, 1999; Traünmüller & Eriksson, 2000). Salient acoustic parameters that are associated with increased vocal effort include increases in fundamental frequency ( $f_0$ ), intensity, duration, and changes in formant frequencies (Eriksson & Traünmüller, 2002; Fux, Feng, & Zimpfer, 2011; Liénard & Benedetto, 1999; Pelegrín-García et al., 2011; Thompson & Webster, 1964; Traünmüller & Eriksson, 2000). The effects of TLD are consistent across talkers, listeners and utterances.

The effect of TLD on  $f_0$  is of particular interest when studying a tone language. If  $f_0$  changes as a function of TLD, will that affect tone perception for native (L1) and second language (L2) listeners? If so, what are the changes that have an impact on tone perception?

Mandarin Chinese is a lexical tone language, and the  $f_0$  height and contour over the domain of a syllable is the primary acoustic cue that differentiates among lexical tones. There are four lexical tones in Mandarin, traditionally referred to as tone 1 to tone 4. For example, the syllable *ma1*, with tone 1, a high-level tone, means “mother”; *ma2* with tone 2, a rising tone, means “hemp”; *ma3* with tone 3, a low falling or low falling-rising tone, means “horse”; and *ma4* with tone 4, a falling tone, means “to scold.” In general, changing the tone of a monosyllabic word alters the meaning of the word, though there are nuances behind this statement worth pointing out. First, there exist many syllable-tone combinations with multiple homophones for which meaning disambiguation is based on writing rather than sound. Second, context in continuous speech helps with the recovery of meaning even if the tone is unclear or wrong (Levow, 2005, 2006). That said, the functional load of tone in explaining phonological contrast in Mandarin is still as high as the functional load of vowels (Surendran & Levow, 2004); hence the role of  $f_0$  in Mandarin speech communication cannot be ignored.

While it is generally agreed that relative  $f_0$  height and  $f_0$  shapes are the primary acoustic cues for Mandarin tones (Chao, 1968; Liu, 1924), many acoustic cues are present simultaneously that may affect the perception of tonal categories; these include contrasts in duration, intensity, creakiness,  $f_0$  height, and  $f_0$  turning point (Blicher, Diehl, & Cohen, 1990; Fu, Zeng, Shannon, & Soli, 1998; Gandour, 1983; Garding, Kratochvil, Svantesson, & Zhang, 1986; Hallé, Chang, & Best, 2004; Lee, Tao, & Bond, 2008; Luo & Fu, 2004; Moore, 1995; Shen & Lin, 1991; Shih, 1988; Shih & Ao, 1996; Wang & Li, 1967; Whalen & Xu, 1992). There is also considerable variation in the  $f_0$  dimensions of natural speech. The  $f_0$  contours of tones produced with contrastive focus and in emphatic speech may look distorted when compared to tones produced in normal speech modes, but they tend to be more separated in tone space, which actually helps with tone recognition (Surendran, 2007; Surendran, Levow, & Xu, 2005; Xu, 1999). Tones produced in prosodically weak positions in connected speech can be reduced to the extent that they lose their distinctive tone shapes (Kochanski & Shih, 2003; Kochanski, Shih, & Jing, 2003; Surendran et al., 2005). The acoustic correlates of tones are further influenced by factors including but not limited to speaker, context, sentence prosody, declination, speaking rate and speaker effort (Kochanski et al., 2003; Shih, 1988, 2000; Xu, 1997, 1999).

Tone learning is an important yet challenging task for second language (L2) learners, who need to recognize the characteristic acoustic attributes that distinguish one tone from another. This task is challenging for speakers of non-tone languages in which the typical function of  $f_0$  is to convey speech act, discourse and paralinguistic information. The functional divergence of  $f_0$  usage is revealed in the different patterns of pitch-shift responses observed in speakers with or without tone language backgrounds (Ning, 2014), as well as the observation that people with musical training are better at tone identification even without prior tone training (Lee & Hung, 2008; Ning, 2014). Making the switch to using  $f_0$  to encode lexical information requires the recruitment of different cortical areas (Gandour et al., 2000; Kaan, Barkley, Bao, & Wayland, 2008; Klein, Zatorre, Milner, & Zhao, 2001; Wang, Spence, Jongman, & Sereno, 1999; Wang, Sereno, Jongman, & Hirsch, 2003). Native speakers of a tone language such as Mandarin are sensitive to tonal contours, while native speakers of American English without tone training pay attention to pitch levels (Huang & Johnson, 2010; Stageray & Downs, 1993), beginning and end points, and the similarity between toneshapes (Gottfried & Suiter, 1997; Kiriloff, 1969; So & Best, 2010; Yang, 2010). Tone 2 and tone 3 have similar shapes and are thus difficult to differentiate. Tone 3 is difficult even for Mandarin-speaking young children to identify (Wong, Schwartz, & Jenkins, 2005).

Having access to multiple examples, especially speech materials rich in variation, allows learners to evaluate the relative contributions multiple cues make to a sound category and to identify the primary cues, thus facilitating learning (Bradlow, Pisoni, Yamada, & Tohkura, 1995; Francis, Ciocca, Ma, & Fenn, 2008; Lively, Logan, & Pisoni, 1993; Lively, Pisoni, Yamada, Tohkura, & Yamada, 1994; Logan, Lively, & Pisoni, 1991; McCandliss, Fiez, Protopapas, Conway, & McClelland, 2002; Wang et al., 1999). Manipulating TLD is an effective method for collecting natural variations in tone. It has been shown that TLD correlates with changes in  $f_0$  minima,  $f_0$  maxima and the initial and final slopes of  $f_0$  contours (Fux et al., 2011). Speech variations induced by TLD have been employed successfully in the tone training of L2 learners (Shih, Lu, Sun, Huang, & Packard, 2010; Shih & Lu, 2010).

The current study builds on previous research on TLD and investigates the effects of TLD on Mandarin Chinese tones through both production and perception experiments. Studying TLD effects on tone will potentially provide a dynamic view of how acoustic attributes combine to account for robust tone classification extending from normal to demanding speech and to identify areas of weakness that arise if tone perception is impacted negatively by TLD. This line of research on the effects of TLD is similar to research on many other modes of speech that are produced under demanding or suboptimal conditions that may impair speech communication and prompt speakers to make adjustments to enhance speech, including the Lombard effect, in which speakers produce clear speech in a noisy environment (Bradlow & Alexander, 2007; Picheny, Durlach, & Braid, 1986, 1989). Clear speech is found to be more intelligible, suggesting that the enhancement measures adopted by speakers are usually effective (Summers, Pisoni, Bernacki, Pedlow, & Stokes, 1988). Nonetheless, L2 learners may have difficulty decoding the altered acoustic cues and thus may not get the full benefits of clear speech (Bradlow & Bent, 2002).

Our research is guided by the following questions:

1. What are the effects of TLD on Mandarin speech?  
Given the robustness of TLD effects reported in the previous literature, our hypothesis is that speech intensity, duration and  $f_0$  height will increase along with TLD.
2. What are the effects of TLD on Mandarin tone ?  
Prior research on TLD suggests a robust effect on the increase in high  $f_0$ . Prior research on the effects of emphatic focus on Mandarin tone suggests a robust effect on the expansion of pitch range, with a clear effect of raising high tones but an inconclusive effect on the behavior of low tones. Our hypothesis is that effects of TLD will dominate, and thus the  $f_0$  of both high and low tones will increase.  
With regard to tone shape, our hypothesis is that the tone shapes corresponding to farther distances will be exaggerated, with high contrast, and those corresponding to close distances will be reduced, with low contrast (Kochanski & Shih, 2003; Kochanski et al., 2003). The acoustic landmarks representing tones at farther TLDs include high  $f_0$ , delayed targets and steeper slopes (Fux et al., 2011; Xu & Wang, 2001; Xu, Xu, & Luo, 1999).
3. What are the effects of TLD on Mandarin tone perception by L1 listeners?  
The goal of TLD-oriented speech is to facilitate speech communication. Our hypothesis is that native speakers and listeners share knowledge of lexical contrasts and have the experience to adapt to changes. We predict that native speakers will be able to make acoustic adjustments within bounds and maintain lexical contrast in adverse conditions, and native listeners will be able to decode the changed signals. Tone perception by native listeners will remain robust under changes in TLD within the scope of the experimental setting.
4. What are the effects of TLD on Mandarin tone perception by L2 listeners?  
Based on prior reports that L2 listeners are unable to reap the full benefits of clear speech, our hypothesis is that the tone recognition rates of low-performing L2 listeners will degrade due to an increase in variation in TLD-oriented speech. As their skill improves, L2 listeners will increasingly reap the benefits of exaggerated tone features.
5. What acoustic attributes affect tone perception by L2 listeners?  
Prior research shows that Mandarin L1 listeners are sensitive to tone contours, while English speakers are sensitive to pitch height (Huang & Johnson, 2010). Tone identification can be more challenging in prosodically reduced positions in which tonal contrast tends to get neutralized (Kochanski et al., 2003; Surendran et al., 2005). Speech produced at a close distance requires less vocal effort, which may have an effect similar to prosodic reduction. Following this line of reasoning, we hypothesize that tones with similar contours, such as Mandarin tones 2 and 3, will be difficult for L2 listeners to differentiate at close talker–listener distances.
6. How can we measure the effects of TLD on tone ?  
Measurement is a methodological question that needs to be addressed. We expect that there is important information in the whole  $f_0$  contour, so we will use tone coefficients obtained from quadratic decomposition to represent the  $f_0$  contour. Acoustic landmarks such as the maximum  $f_0$  will be used in conjunction with tone coefficients.

To answer these research questions, the current work extends TLD research to Mandarin Chinese with both production and perception studies and includes both L1 and L2 listeners in the perception study. This work investigates the relationship between speech production and perception when acoustic attributes change naturally in response to the demands of speech communication.

Sections 2, 3 and 4 report the experimental design, analysis and discussion of the production study, respectively. Sections 5 and 6 report the experimental design and analysis of the perception study, followed by a general discussion and conclusion.

## 2. The production experiment

The goal of the production experiment was to investigate the effects of TLD on tone production. We recorded three speakers saying monosyllabic words with Mandarin lexical tone contrasts at 11 levels of TLD and conducted an acoustic analysis to identify corresponding changes in intensity, duration,  $f_0$  height and  $f_0$  shape. A perception experiment (Section 5) followed the production experiment to investigate whether TLD affects listeners' ability to identify lexical tones.

### 2.1. Experiment design

#### 2.1.1. Speakers

Three native speakers of Mandarin Chinese, including 1 male and 2 females (the authors), participated in the speech recording task. The speech was originally recorded for the purpose of L2 pronunciation training (Shih et al., 2010). All three speakers resided in Illinois at the time of recording. The male speaker came from Beijing and the two female speakers came from Taiwan.

#### 2.1.2. Materials

The materials consisted of recordings of Mandarin monosyllabic words arranged into 15 minimal sets with 4-way tonal contrasts: *ai, cai, chuan, e, guo, ju, ma, qiao, shu, tong, wang, wei, wu, ya* and *yin*. These syllables include a range of Mandarin consonants, vowels and syllable structures and are real words when combined with any of the tones.

Each syllable was recorded by each speaker at 11 different TLDs, repeated 4 times. The crossed design yielded 7920 TLD-oriented tone tokens (11 distances  $\times$  4 tones  $\times$  15 syllables  $\times$  3 speakers  $\times$  4 repetitions).

### 2.1.3. TLD recording procedure

During the recording session, the speaker sat in a soundproofed room with a glass panel in the UIUC Phonetics Lab. A computer monitor was placed 0.46 m in front of the speaker. A Shure 58 microphone was placed on a microphone stand and connected to the computer using a Radio Shack 274-016 XLR jack to 1/4 in. plug adapter/transformer. The microphone was positioned beside the monitor.

A listener and an experimenter were inside the lab but outside the soundproofed room. The experimenter wore a headphone to monitor the recorded speech but sat facing away from the speaker to avoid accidentally making eye contact that could potentially change the TLD control. The listener stood outside of the recording room at one of the 11 distances pre-marked on the floor. For the closest distance level, the listener stood outside the glass panel of the recording room, 0.91 m away from the speaker. Each increasing distance level was 0.51 m farther away. The farthest distance level was at the entry door of the lab, 5.99 m away from the speaker. In this setting, all TLDs were presented with the ambiance of an indoor classroom setting. Table 1 lists TLD levels used in the experiment with the distance expressed in meters.

Recording prompts for the same TLD were recorded in one block in a randomized order. The recording order of TLD levels was also randomized. The experimenter ran a randomization program to select a TLD to record. The listener moved into the designated position, then the recording of the block began. Recording prompts were displayed on a computer screen in the Chinese character set native to the speaker, with traditional characters for speakers from Taiwan and simplified characters for speakers from Mainland China. The speaker was instructed to read the prompt from the monitor first, then speak to the listener while making eye contact. The listener, in turn, was instructed to make eye contact with the speaker. Even though the listener could not hear the speech, as the speaker was in the sound-attenuated booth, the speech gesture was visible, and the listener smiled and nodded to acknowledge each spoken prompt and to facilitate the communication goals of speech under varying TLD.

The speaker controlled the *record* and *stop* buttons via software and could go back to repeat a prompt if he/she chose to do so. The experimenter kept a log of recording errors and at the end of the recording session asked the speaker to reenact TLD levels for individual files.

The distance between the microphone and the speaker as well as the recording volume were kept constant throughout the recording session for each speaker. The volume setting was tested before the recording session, with the listener standing at the farthest TLD.

There were 7986 recorded files, and they were screened by the authors to exclude recording errors and speech errors. The following cases were excluded: files with segmental errors, files with disfluency, files with tone errors that the speaker corrected, and files with recording problems. Most of the recording problems were due to incomplete files. Files were also checked for clipping, though none had perceptible enough clipping that they needed to be excluded. Twenty-seven of the 7986 files were excluded from the recording. There were 16 missing files and 55 duplicates, yielding a total of 7959 files that were kept in the database.

## 2.2. Quadratic decomposition of $F_0$ curves

The effects of TLD on the production and perception of tones may be manifested in information contained in the whole  $f_0$  curve rather than preselected points. Using polynomial decomposition of the  $f_0$  curve is an effective way to address this problem in that it allows the contour to be approximated by a mathematical function. The coefficients of the function can be analyzed in the same way as other numerical representations.

There is a long tradition in the community of speech technology and phonetic research of using polynomial decomposition to represent  $f_0$  curves. Previous research (Chen & Wang, 1990) has used quantized coefficients from polynomial decomposition to represent variations in Mandarin tone and intonation in continuous speech. Machine learning methods have been used to analyze a speech corpus to learn the associations between these tonal coefficients and properties of text that might affect tone shapes in speech, including the preceding and following tonal context, the syllable and word position in the sentences and the length of the sentence. This information has been integrated with a text-to-speech system to predict appropriate tone variations and to generate natural-sounding tones and intonations.

Polynomial equations were used in Andruski and Costello (2004) to model three lexical tones in Green Mong (Hmong) with different phonation types. The researchers successfully classified tones with similar low-falling pitch contours using these coefficients.

In Grabe, Kochanski, and Coleman (2007), dialectal variations in sentence intonation in seven dialects of the British Isles were represented with polynomial coefficients. This work linked linguistic labels of intonational characteristics to mathematical representations. A complex sentence-level  $f_0$  contour was described with a small set of coefficients that were treated as points in

**Table 1**  
TLD levels used in the experiment with distance levels specified in meters.

|          |       |       |       |        |        |       |
|----------|-------|-------|-------|--------|--------|-------|
| Distance | TLD 1 | TLD 2 | TLD 3 | TLD 4  | TLD 5  | TLD 6 |
| Meters   | 0.91  | 1.43  | 1.93  | 2.43   | 2.94   | 3.45  |
| Distance | TLD 7 | TLD 8 | TLD 9 | TLD 10 | TLD 11 |       |
| Meters   | 3.96  | 4.47  | 4.97  | 5.48   | 5.99   |       |



a multidimensional space, making it easy to classify intonation contours on a numerical basis.

An  $f_0$  curve can be approximated by a polynomial function in the form of Eq. (1). In the case of a simple  $f_0$  contour with at most one turning point, the contour can be approximated by a quadratic function, a second-degree polynomial function, as in Eq. (2), which describes the shape of a parabola

$$y = a_0 + a_1x + a_2x^2 + \dots + a_{n-1}x^{n-1} + a_nx^n \quad (1)$$

$$y = a + bx + cx^2 \quad (2)$$

Fig. 1 illustrates how the coefficients  $a$ ,  $b$  and  $c$  in Eq. (2) can be varied to fit curves that approximate monosyllabic Mandarin tones. Because time and  $f_0$  values are positive, the section of the parabola used for tone modeling is in the first quadrant of the Cartesian coordinate system.

Coefficient  $a$  (coef- $a$ ) is a constant. It is the  $y$ -intercept of the parabola and models the initial  $f_0$  height. The effect of changing  $a$  is demonstrated in the leftmost panel of Fig. 1, where the coefficients  $b$  and  $c$  are held constant at  $b=0$  and  $c=0.05$  while  $a$  varies, with the black line showing  $a=100$ , dark gray showing  $a=120$  and light gray showing  $a=80$ . The solid lines show the part of the parabola with positive  $x$  values – that is, the part used to fit a  $f_0$  curve. The dashed lines in the negative range of  $x$  serve only to show what the parabola looks like. When applied to fit  $f_0$  data, coefficient  $a$  is shifted up or down to fit pitch curves. It increases as the initial pitch value increases when everything else is held constant.

Coefficient  $b$  (coef- $b$ ) determines the position of the vertex of the parabola and models the turning point of the tonal contour. The middle panel of Fig. 1 demonstrates the effect of changing  $b$  when  $a$  and  $c$  are held constant. The black line shows that when  $b$  is 0, the parabola is symmetrical with respect to the line  $x=0$ . The dark gray line shows  $b=2$  while the light gray line shows  $b=-2$ . When  $b$  is positive, the vertex of the parabola shifts toward the left, and when  $b$  is negative, the vertex of the parabola shifts toward the right. The intercept is fixed at  $a$ , so the  $y$  value of the vertex changes accordingly. When used to describe  $f_0$  data plotted as a function of time, the relevant part of the curve is to the right of  $x=0$  because time has no negative value. Therefore  $b$  is effective for describing rising curves with different proportions and magnitudes of initial falls, and falling curves with different proportions and magnitudes of initial rises. In the former case,  $b$  can be used to distinguish between two types of rising curves represented by tone 2 and tone 3.

Coefficient  $c$  (coef- $c$ ) models the curvature of the tonal contour. Its effect is demonstrated in the rightmost panel of Fig. 1, where  $a$  and  $b$  are held constant and  $c$  varies. The black line shows  $c=0.1$ ; the dark gray line shows  $c=0.05$ ; and the two light gray lines show  $c=0.01$  and  $c=-0.01$ . As the value of  $c$  approaches 0, the curve becomes wider and shallower. A positive value of  $c$  describes a parabola that opens upward, as in Mandarin tone 2, while a negative value describes a parabola that opens downward, as in Mandarin tone 4.

If  $c$  is 0, then the  $cx^2$  term in Eq. (2) is 0. The equation will then be a linear function, not a quadratic function. When applied to  $f_0$  data, a level tone shape as in Mandarin tone 1 or straight falling tone 4 is expected to have a value of  $c$  near zero. Conversely, a contour tone with a turning point is expected to have a larger magnitude of  $c$ . Typically, a positive value of  $c$  fits a rising tone, while a negative value fits a falling tone. A larger absolute value describes a sharp rise or fall. A smaller absolute value describes a wider curve.

A quadratic function with three coefficients, as in Eq. (2), should be sufficient to describe monosyllabic tones in Mandarin. A polynomial function with more coefficients is needed for sentence-level intonation with more complex contours.

### 2.3. Semitone conversion

In the current study,  $f_0$  values in Hz were converted to semitones (ST) before polynomial decomposition following Eq. (3). Given the logarithmic nature of the semitone scale, it is one of the most effective scales for representing the psychoacoustic aspects of pitch (Nolan, 2003; Trautmann & Eriksson, 2000). In Eq. (3),  $f_{0a}$  is the  $f_0$  value in Hz to be converted in reference to a base value  $f_{0b}$ ,

$$ST = 12 \cdot \log_2 \frac{f_{0a}}{f_{0b}} \quad (3)$$

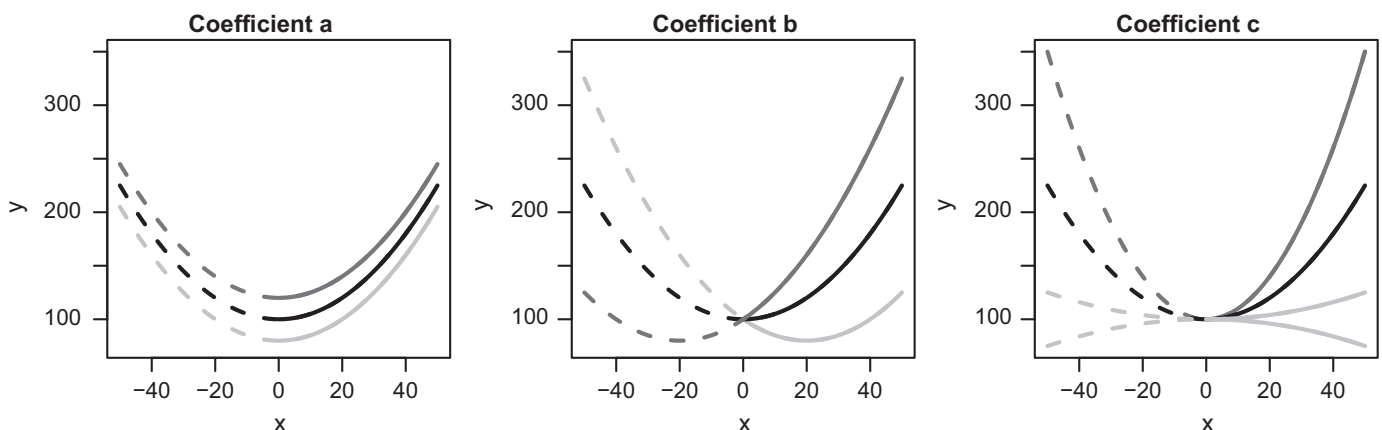


Fig. 1. Plots of the quadratic function showing the effect of changing one coefficient while the others are kept constant.

Fundamental frequency values in Hz from each speaker were analyzed, and the first quartile was used as the base frequency for semitone conversion. The lowest  $f_0$  values were not used for they were typically the result of creaky voice, which often co-occurs with tone 3 due to its low pitch range. The base frequencies for the three speakers were 182.4 Hz, 192.9 Hz and 98.5 Hz for speaker SC (female), SD (female) and SL (male), respectively.

#### 2.4. Data preparation

Speech files were processed with ESPS/Waves by Entropic (Talkin & Lin, 1996) to obtain  $f_0$  tracking, voicing information and RMS energy information at 100 samples per second. The following parameters were extracted from the sound files in order to study the effects of TLD:

- *Duration:*

The syllable duration was calculated from sound segmentations obtained using the Mandarin version of the P2FA forced aligner (Yuan & Liberman, 2008).

- *Intensity:*

The intensity value of each syllable was computed as the area under the intensity curve bounded by the syllable boundaries. The intensity curve in turn was represented by the RMS energy value of the sound wave. Each RMS value was smoothed over a 30 ms Hanning window, with the left edge of the window placed 5 ms before the beginning of the frame (Talkin & Lin, 1996). The square of the intensity value is equal to the acoustic power discussed in the introduction.

- *Tone:*

The  $f_0$  curve was first processed to reduce  $f_0$  effects unrelated to tone with an R script that applied the same criteria to all files. The first effect that had to be reduced was that  $f_0$  samples at the end of a syllable occasionally move in the opposite direction of the tone movement to return to the rest position, creating an extra  $f_0$  turning point that a quadratic equation cannot fit well. In these cases, the RMS energy is low and speech is barely audible. This type of  $f_0$  effect was addressed by removing  $f_0$  samples at the ends of syllables when RMS values were less than 300, about twice the median RMS value of the noise floor. Second, there can be erratic  $f_0$  values at the beginnings of  $f_0$  contours that reflect an unstable stage of the vocal cords as they transition from consonant to vowel. The script checked the first six  $f_0$  samples and eliminated  $f_0$  jumps larger than 15 Hz. Finally, erratic  $f_0$  values in the middle of a tone 3 are common, as the low pitch leads to creakiness when the pitch is too low for the vocal folds to sustain regular vibration. Samples that showed this effect were kept because they represent a realistic physical phenomenon with implications for perception.

The  $f_0$  curve was then smoothed and converted to semitones following Eq. (3). Each speaker's first quartile  $f_0$  value was used as the base. The resulting  $f_0$  curve was scaled to 50 samples per syllable, and then fitted with the quadratic function to obtain three quadratic coefficients (coef-a, coef-b, coef-c) to represent the tonal contour.

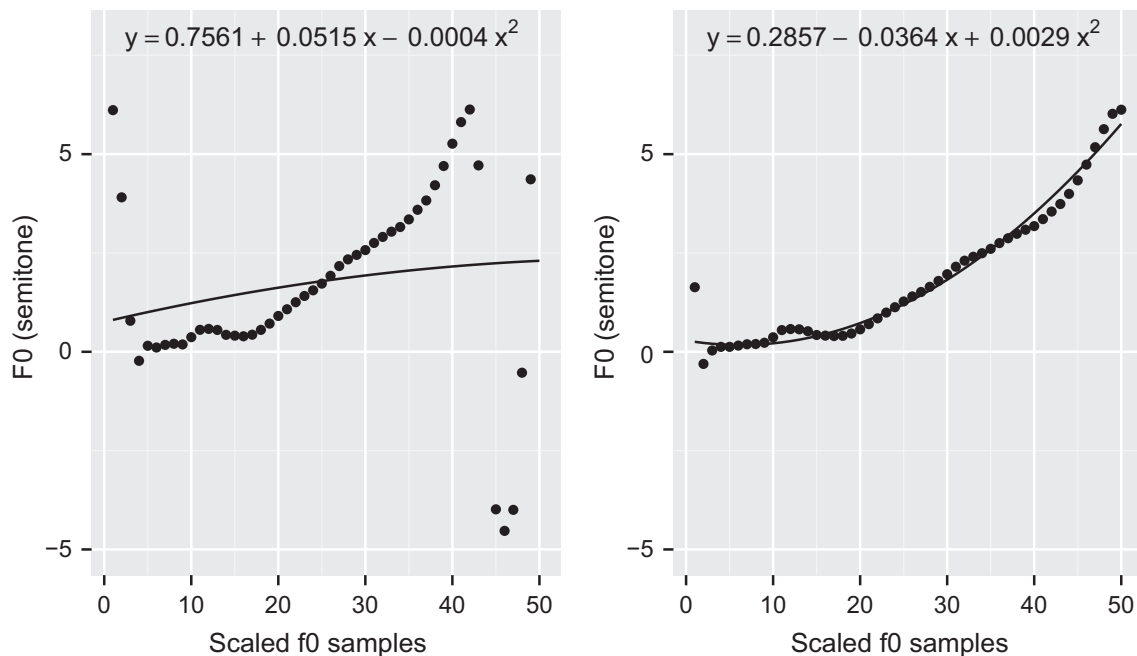


Fig. 2. An example using quadratic equations to fit  $f_0$  curves. The solid circles are  $f_0$  data in semitones. The black solid line is the quadratic equation given in each panel that fits the curve. The left panel shows an  $f_0$  contour with segmental effects, which affect the quadratic fit. The right panel shows an  $f_0$  contour with reduced segmental effects in which the quadratic equation fits the tone shape.

Fig. 2 presents an example that shows how a quadratic equation is used to represent a  $f_0$  curve and how systematic trimming of segmental effects is beneficial to obtaining a good fit. The speech file is for the syllable *wu2* (tone 2) produced by the male speaker, SL, at TLD level 3. The solid circles are the  $f_0$  data in semitones. The black solid lines are the quadratic functions that fit the curves. The left panel plots the  $f_0$  data obtained from the syllable boundaries with the voiceless region excluded. The high  $f_0$  samples at the beginning and low ones at the end affect the quadratic fit strongly, and the result,  $0.7561 + 0.0515x - 0.0004x^2$ , is not representative of tone production. The right panel plots the trimmed  $f_0$  data with most  $f_0$  jumps at the beginning and low intensity points at the end excluded. The resulting quadratic equation,  $0.2857 - 0.0364x + 0.0029x^2$ , describes the  $f_0$  contour reasonably well.

The average root-mean-square deviation between the parameterized  $f_0$  contour and the actual contour for the whole corpus is 5.89 Hz, with the first and third quartiles being 2.86 Hz and 6.94 Hz, respectively. When calculated in semitones, the average deviation is 0.58 semitones, with first and third quartiles being 0.25 and 0.62 semitones, respectively.

### 3. Analysis and results

This section presents analyses of the acoustic attributes of responses to TLD, including intensity, duration and the acoustic correlates of tone. Descriptive analyses are presented first, followed by results of statistical analyses.

In addition to the main fixed effects TLD and tone, the experiment design included three sources of random effects: speaker, syllable and repetition. Based on Pinheiro and Bates (2000), Baayen (2008), Wieling, Nerbonne, and Baayen (2011), Weiss (2010), Winter (2013) and references cited therein, mixed effects regression models are effective for modeling multiple random effects. A linear regression model assumes that data points are independent observations. However, repeated measures, data produced by the same speaker or for the same syllable are correlated, therefore violating the assumption of independence. The linear mixed effects regression models correct data correlation problems due to random effects and thus allows more robust estimation of fixed effects. The clean separation of fixed and random effects is a highly desirable feature in speech synthesis, making it possible to build a speaker-independent model using fixed effects and then adapt it to speaker-dependent systems by adding speaker coefficients from the random effects.

We ran a series of mixed effect regression models using R packages lme4 (Bates, Maechler, Bolker, & Walker, 2014) and nlme (Pinheiro, Bates, DebRoy, & Sarkar, 2013) to evaluate the influence of fixed and random effects on each of the following acoustic measures: log intensity, duration, maximum  $f_0$ , and the three tone coefficients, coef-a, coef-b and coef-c. In each model, one of the acoustic measures was the dependent variable. The goal was to find out how the acoustic attribute in question changes along with TLD while taking into account effects of other variables.

In addition to descriptive and statistical analyses, principal component analysis (PCA) was conducted to find a linear combinations of variables that best explain the data in order to reduce the dimensionality of correlated acoustic measures. Classification and Regression Trees (CART) (Breiman, Friedman, Olshen, & Stone, 1984) were used to explore the effects of TLD on tone classification. All data analyses in this Section were performed using R (R Core Team, 2012).

#### 3.1. Log intensity

Fig. 3 presents the density plot of log intensity by TLD, where data from all speakers and all tones are combined. Intensity was calculated as the area under the RMS energy curve of the speech waveform. Applying the logarithm function transformed the data to a normal distribution. There is a monotonic increase in log intensity plotted on the x-axis, reflecting the effects of different TLD levels, which are coded by increasingly lighter shades of gray.

After log transform, the relationship between intensity and TLD was linear for TLD levels tested within the scope of the experiment. There is a strong positive correlation between TLD and log intensity ( $r=0.71$ ,  $t=90.39$ ,  $df=7957$ ,  $p<0.0001$ ). Table 2 presents the mean, median and standard deviation of the log intensity values by TLD, confirming with numerical data the monotonic increase observed in the density plot. Data from different TLD levels are presented in separate rows. The mean and median log intensity values increase along with TLD. The similarity between mean and median values of the same TLD and the similar values in standard deviation for different TLDs indicate a similar pattern of variation in log intensity across TLD conditions. The speakers were making effort to compensate for the loss of sound intensity, which obeys the inverse-square law, though the compensation is partial.

The log intensity data was analyzed using linear mixed effects regression models. The selection of the model started with a visual inspection of the distribution of the data and interaction plots, which was followed by a procedure of model building and comparison with Bonferroni correction (Baayen, 2008; Singer & Willett, 2003; Weiss, 2010) using measures including Akaike's information criterion (AIC), the Bayesian information criterion (BIC) and the log-likelihood ratio (LogLik), which estimates the relative quality of a model. The AIC and BIC reveal overfitting by evaluating the goodness-of-fit against the complexity of the model, penalizing it for the number of parameters being estimated. For AIC and BIC, smaller numbers indicate a better model, whereas for LogLik, a higher number indicates a better model. The numbers are used to compare models fitted with the same data and are not comparable across datasets.

Linear mixed effects regression models do not report significance in  $p$ -values in the traditional way. In this study, the significance levels of fixed effects were computed with a conservative estimate of the degree of freedom (Baayen, 2008). The significance levels of random effects were evaluated by using an ANOVA with Bonferroni correction to test whether two models built with and without a term were significantly different.

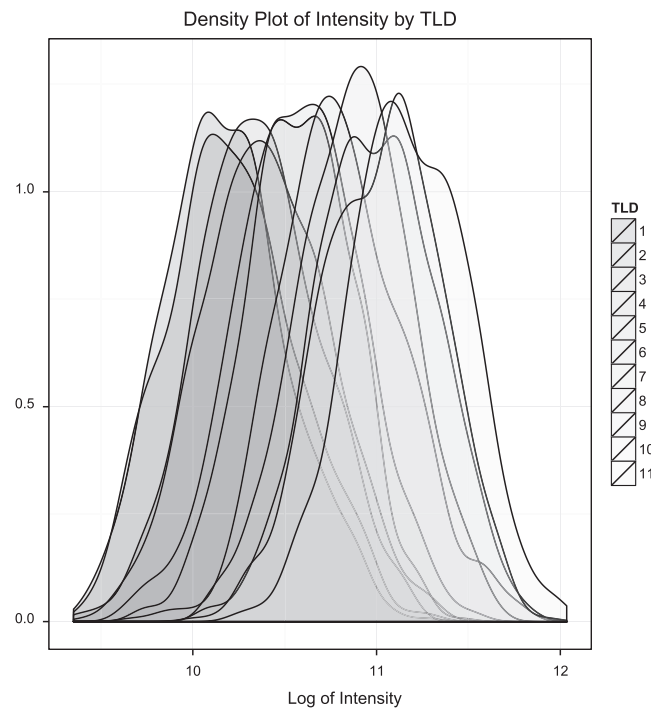


Fig. 3. Density plot of log intensity by TLD. There is a monotonic increase in log intensity plotted on the x-axis reflecting increasing TLD levels, coded by increasing lighter shades of gray.

Table 2

Mean, median and standard deviation of log intensity by TLD. Data from different TLD levels are presented in separate rows.

|        | Mean  | Median | SD    |
|--------|-------|--------|-------|
| TLD 1  | 10.18 | 10.16  | 0.325 |
| TLD 2  | 10.22 | 10.21  | 0.345 |
| TLD 3  | 10.36 | 10.35  | 0.327 |
| TLD 4  | 10.40 | 10.38  | 0.339 |
| TLD 5  | 10.54 | 10.54  | 0.309 |
| TLD 6  | 10.61 | 10.61  | 0.309 |
| TLD 7  | 10.81 | 10.79  | 0.334 |
| TLD 8  | 10.88 | 10.89  | 0.301 |
| TLD 9  | 11.00 | 11.00  | 0.318 |
| TLD 10 | 11.03 | 11.06  | 0.317 |
| TLD 11 | 11.18 | 11.16  | 0.308 |

Table 3

Summary of the model predicting log intensity from fixed effects TLD, tone, the interaction of TLD and tone, and random intercepts for repetition, syllable and speaker crossed with tone.

|                 | AIC<br>-170 | BIC<br>-86  | logLik<br>97 | Deviance<br>-194 | df.resid<br>7947 |
|-----------------|-------------|-------------|--------------|------------------|------------------|
| Random effects: | Groups      | Name        |              | Std.Dev.         |                  |
|                 | syll        | (Intercept) |              | 0.10             |                  |
|                 | spk:tone    | (Intercept) |              | 0.14             |                  |
|                 | rep         | (Intercept) |              | 0.06             |                  |
|                 | Residual    |             |              | 0.24             |                  |
| Fixed effects:  | Estimate    | Std. error  | t value      | p value          |                  |
| (Intercept)     | 10.32       | 0.087       | 118.24       | <0.0001          |                  |
| TLD             | 0.09        | 0.001       | 54.14        | <0.0001          |                  |
| tone 2          | -0.32       | 0.111       | -2.90        | <0.01            |                  |
| tone 3          | -0.50       | 0.111       | -4.54        | <0.0001          |                  |
| tone 4          | -0.31       | 0.111       | -2.78        | <0.01            |                  |
| TLD:tone 2      | 0.02        | 0.002       | 7.96         | <0.0001          |                  |
| TLD:tone 3      | 0.02        | 0.002       | 9.04         | <0.0001          |                  |
| TLD:tone 4      | 0.01        | 0.002       | 4.77         | <0.0001          |                  |



Eq. (4) gives the R formula of the mixed effects model selected through a round of model competition to analyze the log intensity data. The model predicted log intensity data using TLD levels, tone, and the interaction of TLD and tone as the fixed effects. The random effects model included a random intercept for each level of syllable and repetition as well as speaker crossed with tone. We took advantage of the linear relationship between log intensity and TLD and used TLD as a numerical variable in the model. Tone was used as a categorical variable. In this model as well as other models in this section, the number of observations is 7959. Groups include syllable (15 levels), speaker (3 levels), tone (4 levels) and repetition (4 levels). The model explained 74% of the variance in the data.

$$\log(\text{intensity}) \sim \text{TLD} * \text{tone} + (1 \mid \text{spk} : \text{tone}) + (1 \mid \text{rep}) + (1 \mid \text{syll}) + e \quad (4)$$

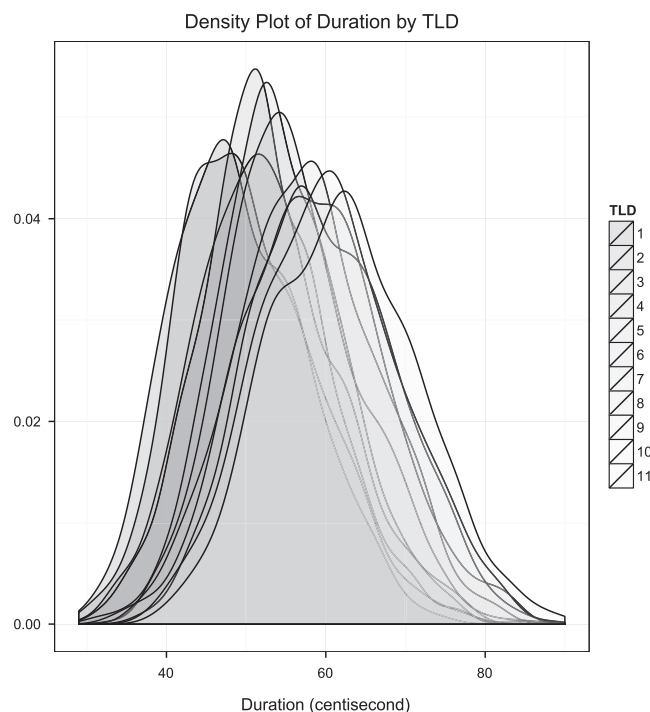
Tables 3, 4 and 5 summarize the results. Repetition effects are described in text. In Table 3, the intercept of the fixed effects was 10.32, and each level of increase in TLD added 0.09. The tone factor in the fixed effects showed negative coefficients for tones 2, 3 and 4 relative to tone 1, suggesting that these tones were produced with a softer voice than tone 1. The coefficients for the interaction of TLD and tone all had values that were positive with the effect of increasing log intensity by tone and by TLD. The effects of tone 2 and tone 4 were significant at the level of  $p < 0.01$ . All other fixed effects, including the TLD effect, the effect of tone 3, and the interaction of TLD and tone, were significant at the level of  $p < 0.0001$ .

**Table 4**  
Random intercepts adjusting the value of log intensity by speaker crossed with tone.

|             | Tone 1 | Tone 2 | Tone 3 | Tone 4 |
|-------------|--------|--------|--------|--------|
| SC (female) | −0.078 | −0.049 | −0.297 | −0.080 |
| SD (female) | −0.064 | 0.056  | 0.086  | −0.064 |
| SL (male)   | 0.142  | −0.007 | 0.211  | 0.145  |

**Table 5**  
Random intercepts adjusting the value of log intensity by syllable.

|                  |       |       |       |       |       |
|------------------|-------|-------|-------|-------|-------|
| Syllable         | ju    | shu   | wei   | e     | tong  |
| Random intercept | −0.17 | −0.11 | −0.10 | −0.07 | −0.05 |
| Syllable         | yin   | wu    | cai   | chuan | qiao  |
| Random intercept | −0.05 | −0.04 | −0.04 | −0.01 | 0.05  |
| Syllable         | ai    | ya    | guo   | ma    | wang  |
| Random intercept | 0.07  | 0.10  | 0.12  | 0.15  | 0.16  |



**Fig. 4.** Density plot of duration by TLD. Duration (in centiseconds) plotted on the x-axis lengthens with the increase in TLD levels, which are coded by increasingly lighter shades of gray.

The predicted log intensity values were adjusted in the model with random intercepts fitted for repetition, syllable and speaker by tone. The random intercepts of repetition were  $-0.078$ ,  $0.003$ ,  $0.060$  and  $0.015$  for repetitions 1–4, respectively. Table 4 presents random intercepts adjusting log intensity by speaker crossed with tone. The male speaker is louder than the female speakers in most cases, as expected. Table 5 presents the random intercepts adjusting the log intensity values by syllable. The scale is consistent with the expectations of articulatory phonetics and acoustics, in that low vowels are louder while high vowels are softer.

### 3.2. Duration

Duration also increases with TLD as shown in Fig. 4, which plots the density function of duration by TLD, and Table 6, which presents the mean, median and standard deviation of duration in centiseconds. In general, duration lengthens as TLD increases, though the pattern of monotonicity is not as strong as in the case of log intensity, with some adjacent TLD levels having similar durations. There is a moderate positive correlation between TLD and duration ( $r = 0.43$ ,  $t = 42.63$ ,  $df = 7957$ ,  $p < 0.0001$ ).

The building and selection of the duration model followed the same procedure as that of the log intensity model. Tables 7, 8 and 9 present a summary of the results from the final linear mixed effects model. Talker-to-listener distance, tone, and their interaction term were included in the fixed effects model. The random effects model included random intercepts of syllable, repetition and speaker crossed with tone. The model explained 78% of the variance in the data.

The intercept of the fixed effects was 49.03 centiseconds. For each increasing level of TLD, duration increased by 1.09 centiseconds. The TLD effect was significant at the level of  $p < 0.001$ . The fixed effects coefficient of tone 4 was significant at the level of  $p < 0.05$ , while the effects of tone 2 and tone 3 were not significant, as the burden of explanation was shifted to the interaction term, where the interaction effects of TLD and tones were all significant.

Tone 3 lengthened duration relative to tone 1, while tone 2 and tone 4 had a shortening effect. Combining the effects of tone and the interaction of tone and TLD, the duration scale by tone was  $\text{tone3} > \text{tone1} > \text{tone2} > \text{tone4}$ .

Table 8 presents the random intercepts adjusting duration by syllable. The syllable duration increases with the number of segments in the syllable. Intrinsically long consonants such as fricatives and aspirated stops/affricates also add to the syllable duration. The effects are consistent with previous studies and models of Mandarin duration (de Jong, 2004; Feng, 1985; House &

**Table 6**

Mean, median and standard deviation of syllable duration in centiseconds. Data from different TLD levels are presented in separate rows.

|        | Mean (cs) | Median (cs) | SD (cs) |
|--------|-----------|-------------|---------|
| TLD 1  | 49.2      | 48          | 8.275   |
| TLD 2  | 50.4      | 49          | 8.530   |
| TLD 3  | 52.3      | 52          | 8.025   |
| TLD 4  | 52.7      | 52          | 8.700   |
| TLD 5  | 54.6      | 54          | 7.912   |
| TLD 6  | 55.8      | 55          | 8.238   |
| TLD 7  | 57.8      | 58          | 8.686   |
| TLD 8  | 58.6      | 58          | 9.025   |
| TLD 9  | 59.7      | 59          | 8.859   |
| TLD 10 | 60.4      | 60          | 8.975   |
| TLD 11 | 62.2      | 62          | 9.173   |

**Table 7**

Summary of the model predicting duration (centiseconds) from fixed effects TLD, tone, and the interaction of TLD and tone, and random effects syllable, repetition and speaker crossed with tone.

|                 | AIC<br>46,754 | BIC<br>46,838 | logLik<br>-23,365 | deviance<br>46,730 | df.resid<br>7947 |
|-----------------|---------------|---------------|-------------------|--------------------|------------------|
| Random effects: | Groups        | Name          |                   | Std.Dev.           |                  |
|                 | syll          | (Intercept)   |                   | 3.585              |                  |
|                 | spk:tone      | (Intercept)   |                   | 3.802              |                  |
|                 | rep           | (Intercept)   |                   | 0.850              |                  |
|                 | Residual      |               |                   | 4.509              |                  |
| Fixed effects:  | Estimate      | Std. error    | <i>t</i> value    | <i>P</i> value     |                  |
| (Intercept)     | 49.03         | 2.430         | 20.18             | <0.001             |                  |
| TLD             | 1.09          | 0.032         | 34.30             | <0.001             |                  |
| tone 2          | -0.75         | 3.119         | -0.24             | 0.81               |                  |
| tone 3          | 4.67          | 3.119         | 1.50              | 0.14               |                  |
| tone 4          | -8.03         | 3.119         | -2.57             | <0.05              |                  |
| TLD:tone 2      | 0.15          | 0.045         | 3.30              | <0.01              |                  |
| TLD:tone 3      | 0.49          | 0.045         | 10.83             | <0.001             |                  |
| TLD:tone 4      | 0.18          | 0.045         | 3.91              | <0.001             |                  |

**Table 8**

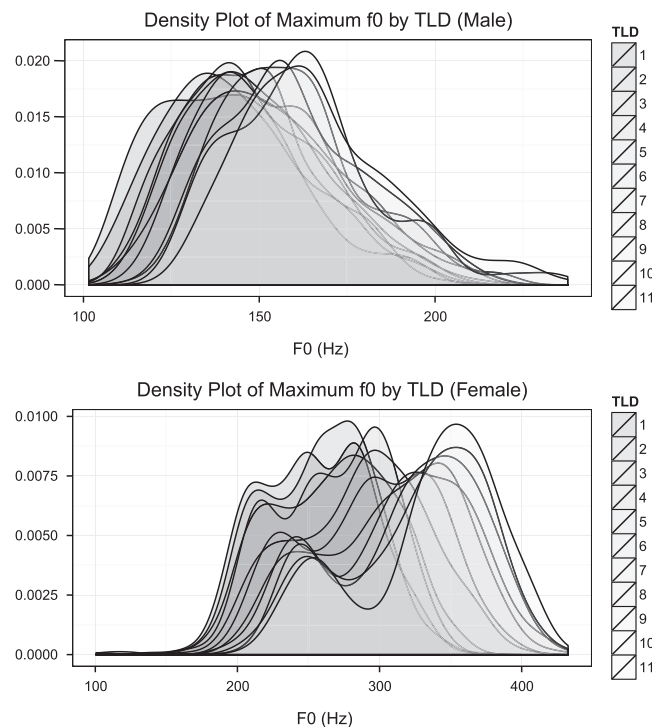
Random intercepts adjusting duration by syllable. The unit is centiseconds.

|                  |       |       |       |       |       |
|------------------|-------|-------|-------|-------|-------|
| Syllable         | e     | wu    | ya    | ma    | ai    |
| Random intercept | −5.29 | −4.31 | −4.18 | −2.77 | −2.07 |
| Syllable         | wei   | yin   | ju    | guo   | wang  |
| Random intercept | −1.97 | −1.45 | −0.48 | −0.01 | 1.11  |
| Syllable         | tong  | qiao  | cai   | shu   | chuan |
| Random intercept | 3.03  | 3.03  | 3.66  | 5.35  | 6.36  |

**Table 9**

Random intercepts adjusting duration by speaker crossed with tone. The unit is centiseconds.

|    | Tone 1 | Tone 2 | Tone 3 | Tone 4 |
|----|--------|--------|--------|--------|
| SC | 4.17   | 4.40   | −2.90  | 3.47   |
| SD | −0.13  | 0.91   | −3.24  | 0.46   |
| SL | −4.04  | −5.31  | 6.14   | −3.93  |

**Fig. 5.** Density plot of maximum  $f_0$  by TLD. The top panel shows data from the male speaker and the bottom panel shows combined data from the two female speakers. The x-axis shows maximum  $f_0$  in Hz separated by TLD, and TLD levels are coded by different shades of gray.

Fairbanks, 1953; Shih & Ao, 1996; van Santen & Shih, 2000). The random intercepts of repetition were 0.442, 0.226, 0.595 and −1.263 for repetitions 1–4, respectively. The random intercepts of speaker by tone are presented in Table 9.

### 3.3. Maximum $f_0$

As TLD increases, maximum  $f_0$  values tend to increase due to greater vocal effort. Fig. 5 presents the density plots of maximum  $f_0$  by TLD. The top panel shows data from the male speaker and the bottom panel shows data from two female speakers. The bimodal distribution in the female density plot is due to the higher maximum  $f_0$  of tone 1 and tone 4 on one hand, and the lower maximum  $f_0$  of tone 2 and tone 3 on the other. Generally, Fig. 5 shows an increase in maximum  $f_0$  by TLD even though speaker and tone contribute to more data variation.

This section analyze maximum  $f_0$  in Hz to make it possible to compare results of TLD research on  $f_0$  using Hz, and to allow straightforward control of TLD-sensitive speech synthesis, where the  $f_0$  values in Table 10 can be plugged into such systems.

**Table 10**

Mean, median and standard deviation of the maximum  $f_0$  values in Hz. Data from different TLD levels are presented in separate rows, and data from speakers are separated into 3 blocks, with the data from the male speaker in the rightmost block.

|        | SC (female) |        |    | SD (female) |        |    | SL (male) |        |    |
|--------|-------------|--------|----|-------------|--------|----|-----------|--------|----|
|        | Mean        | Median | SD | Mean        | Median | SD | Mean      | Median | SD |
| TLD 1  | 242         | 245    | 34 | 267         | 277    | 41 | 141       | 140    | 21 |
| TLD 2  | 252         | 257    | 37 | 262         | 270    | 38 | 143       | 141    | 21 |
| TLD 3  | 256         | 259    | 38 | 280         | 284    | 42 | 146       | 144    | 19 |
| TLD 4  | 262         | 266    | 38 | 273         | 280    | 40 | 149       | 145    | 20 |
| TLD 5  | 275         | 286    | 41 | 299         | 308    | 48 | 150       | 146    | 20 |
| TLD 6  | 283         | 291    | 42 | 305         | 318    | 50 | 154       | 152    | 22 |
| TLD 7  | 293         | 301    | 44 | 323         | 336    | 49 | 154       | 151    | 21 |
| TLD 8  | 299         | 309    | 46 | 325         | 341    | 53 | 159       | 157    | 20 |
| TLD 9  | 311         | 321    | 47 | 332         | 344    | 50 | 160       | 158    | 21 |
| TLD 10 | 314         | 325    | 49 | 336         | 351    | 48 | 162       | 161    | 22 |
| TLD 11 | 325         | 338    | 51 | 343         | 359    | 48 | 167       | 164    | 22 |

**Table 11**

Summary of the model predicting maximum  $f_0$  (Hz) from fixed effects TLD and tone with an interaction term, and random effects syllable, repetition and speaker crossed with tone.

|                 | AIC      | BIC         | logLik         | Deviance       | df.resid |
|-----------------|----------|-------------|----------------|----------------|----------|
|                 | 69,171   | 69,255      | -34,574        | 69147          | 7947     |
| Random effects: | Groups   | Name        |                | Std. dev.      |          |
|                 | syll     | (Intercept) |                | 5.17           |          |
|                 | spk:tone | (Intercept) |                | 67.669         |          |
|                 | rep      | (Intercept) |                | 3.912          |          |
|                 | Residual |             |                | 18.430         |          |
| Fixed effects:  | Estimate | Std. error  | <i>t</i> value | <i>p</i> value |          |
| (Intercept)     | 220.28   | 39.151      | 5.63           | <0.0001        |          |
| TLD             | 6.97     | 0.13        | 53.55          | <0.0001        |          |
| tone 2          | -25.01   | 55.266      | -0.45          | 0.651          |          |
| tone 3          | -49.17   | 55.266      | -0.89          | 0.374          |          |
| tone 4          | 22.44    | 55.266      | 0.41           | 0.685          |          |
| TLD:tone 2      | 0.54     | 0.184       | 2.94           | <0.01          |          |
| TLD:tone 3      | -2.47    | 0.185       | -13.35         | <0.0001        |          |
| TLD:tone 4      | 0.18     | 0.184       | 0.97           | 0.334          |          |

**Table 12**

Random intercepts adjusting maximum  $f_0$  (Hz) by syllable.

| Syllable         | ai     | cai   | ya    | ma    | e     |
|------------------|--------|-------|-------|-------|-------|
| Random intercept | -10.15 | -5.11 | -4.23 | -4.13 | -4.02 |
| Syllable         | wei    | wang  | qiao  | guo   | ju    |
| Random intercept | -1.78  | -1.13 | 0.61  | 1.12  | 1.81  |
| Syllable         | chuan  | yin   | wu    | tong  | shu   |
| Random intercept | 2.73   | 4.30  | 4.52  | 5.57  | 9.91  |

**Table 13**

Random intercepts adjusting maximum  $f_0$  (Hz) by speaker crossed with tone.

|             | tone 1  | tone 2 | tone 3 | tone 4  |
|-------------|---------|--------|--------|---------|
| SC (female) | 37.08   | 38.69  | 24.26  | 43.96   |
| SD (female) | 67.42   | 56.15  | 42.27  | 63.52   |
| SL (male)   | -104.50 | -95.11 | -66.53 | -107.48 |

Table 10 presents the mean, median and standard deviation of the maximum  $f_0$  in Hz with data separated by TLD level and speaker. Data from different TLD levels are presented in separate rows. Data from the male speaker are in the rightmost block, and data from the two female speakers are in the center and left blocks. There is a weak positive correlation between TLD and maximum  $f_0$  ( $r = 0.257, t = 23.763, df = 2960, p < 0.001$ ), even with gender and tone data combined.

Tables 11–13 present major results from the linear mixed effects model selected through model competition. Talker-to-listener distance and tone were included as fixed effects, and their interaction term was included. Syllable, repetition and speaker crossed with tone were included in the random intercepts models. The model explained 94.7% of the variance in the data.

The intercept of the fixed effects was 220 Hz. For each increased level of TLD, maximum  $f_0$  increased by 7 Hz. Tone 4 had an effect of increasing maximum  $f_0$  by 22 Hz more than tone 1. Tone 2 and tone 3 reduced maximum  $f_0$  by 25 Hz and 49 Hz, respectively, although the tone effects were not significant. The effects of TLD and the interaction of TLD and tone 3 were significant at the level of  $p < 0.0001$ . The interaction of TLD and tone 2 was significant at the level of  $p < 0.01$ .

The predicted maximum  $f_0$  values were further adjusted by random intercepts fitted for repetition, syllable, and speaker crossed with tone. The random intercepts of repetition were  $-3.039, -3.400, 1.804, 4.636$  for repetitions 1–4, respectively. Table 12 presents the random intercepts adjusting maximum  $f_0$  by syllable. The scale is consistent with the prediction of the vowel's intrinsic effect on  $f_0$ : the  $f_0$  value correlates with vowel height, where a low vowel tends to have a lower  $f_0$  and a high vowel tends to have a higher  $f_0$  (House & Fairbanks, 1953; Peterson & Barney, 1952). Table 13 presents random intercepts adjusting maximum  $f_0$  by speaker and tone.

### 3.4. Models of tone coefficients

The  $f_0$  contour of each sound file was converted to semitones, scaled to 50 samples, then fitted with a quadratic function (see Section 2.2). The curve-fitting process produced three numbers, the three coefficients of the quadratic function, that represent the whole tonal contour. Fig. 6 presents three interaction plots of TLD and tone coefficients  $a$ ,  $b$  and  $c$ . Talker-to-listener distance is plotted on the x-axes, and mean coefficient values are plotted on the y-axes. Circles, upward triangles, squares, and plus signs

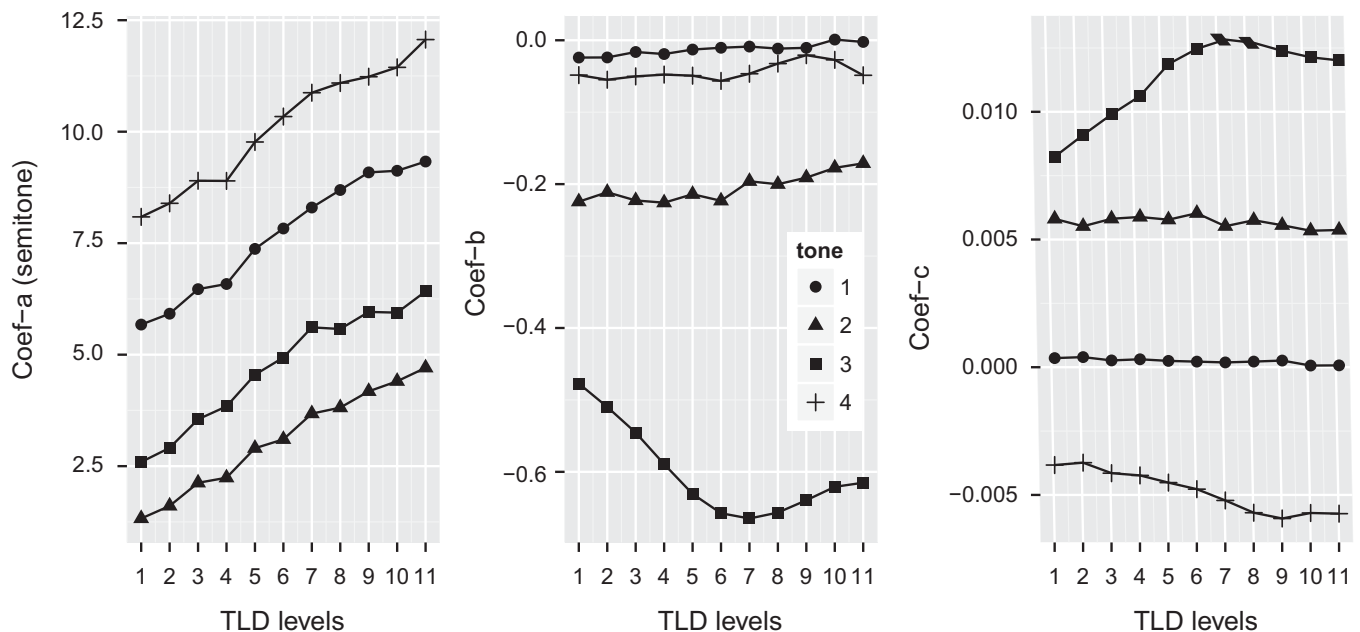


Fig. 6. Interaction plots of TLD and tone coefficients  $a$ ,  $b$  and  $c$ . Talker-to-listener distance is plotted on the x-axes and mean coefficient values are plotted on the y-axes. Circles, upward triangles, squares, and plus signs represent tone 1 to tone 4, respectively.

Table 14

Summary of the model predicting coef-a from fixed effects TLD and tone and random effects syllable, repetition and speaker.

|                 | AIC<br>30,737 | BIC<br>30,800 | logLik<br>-15,360 | deviance<br>30,719 | df.resid<br>7950 |
|-----------------|---------------|---------------|-------------------|--------------------|------------------|
| Random effects: | Groups        | Name          |                   | Std. dev.          |                  |
|                 | syll          | (Intercept)   |                   | 0.596              |                  |
|                 | rep           | (Intercept)   |                   | 0.237              |                  |
|                 | spk           | (Intercept)   |                   | 0.483              |                  |
|                 | Residual      |               |                   | 1.658              |                  |
| Fixed effects:  | Estimate      | Std. error    | $t$ value         | $p$ value          |                  |
| (Intercept)     | 5.349         | 0.344         | 15.56             | <0.0001            |                  |
| TLD             | 0.386         | 0.006         | 65.71             | <0.0001            |                  |
| tone 2          | -4.567        | 0.053         | -87.00            | <0.0001            |                  |
| tone 3          | -2.947        | 0.053         | -56.13            | <0.0001            |                  |
| tone 4          | 2.434         | 0.053         | 46.38             | <0.0001            |                  |



represent tone 1–tone 4. The left panel shows the mean values of *coef-a* as they increase with TLD. The middle panel shows the mean values of *coef-b*, demonstrating an interaction effect with tone and TLD. Coefficient *b* values do not change much with TLD for tone 1 and tone 4 but increase slightly for tone 2 and decrease for tone 3 up to TLD level 7, where they level off. The right panel shows the mean values of *coef-c*, which seem to correlate negatively with *coef-b*. Again, there appears to be an interaction effect with tone by TLD. As TLD increases, the values of *coef-c* increase for tone 3 up to TLD level 7 and then level off. Coefficient *c* does not seem to be affected by TLD for tone 1, while it decreases slightly for tone 2 and tone 4.

Pearson's product-moment correlation coefficient was used to evaluate whether there were linear relationships between pairs of tone coefficients. There was a very weak correlation between *coef-a* and *coef-b* ( $r = -0.10$ ,  $t = 9.1285$ ,  $df = 7957$ ,  $p < 0.0001$ ), a moderate negative linear relationship between *coef-a* and *coef-c* ( $r = -0.38$ ,  $t = -37.07$ ,  $df = 7957$ ,  $p < 0.0001$ ), and a strong negative linear relationship between *coef-b* and *coef-c* ( $r = -0.91$ ,  $t = -195.09$ ,  $df = 7957$ ,  $p < 0.0001$ ). All three cases were statistically significant. We then proceeded to build models to predict tone coefficients.

#### 3.4.1. Tone coefficient *a*

A linear mixed effects model was built following the procedure of model comparison and selection described in Section 3.1. In the final model, TLD and tone were included as fixed effects without an interaction term. Syllable, repetition and speaker were included in the random intercepts models. The model explained 77% of the variance in the data. The results are presented in Tables 14 and 15, with repetition effects and speaker effects described in text.

The intercept of the fixed effects was 5.35 semitones. For each increasing level of TLD, the value of *coef-a* increased by 0.386 semitones. Tone 2 and tone 3 had the effect of decreasing the value of *coef-a* relative to tone 1, while tone 4 had the effect of increasing the value. The TLD effect and the tone effects were all significant at the level of  $p < 0.0001$ .

The significance of the random effects factors was evaluated by model comparison using an ANOVA with Bonferroni correction. All included factors were significant at the  $p < 0.001$  level. The random intercepts for repetition were  $-0.179$ ,  $-0.207$ ,  $0.087$  and  $0.299$  semitones for repetitions 1–4. The random intercepts for speaker were 0.555 for SC (female),  $-0.560$  for SD (female) and 0.005 for SL (male). The gender effect was normalized by semitone conversion. Table 15 presents the random intercepts adjusting *coef-a* by syllable. In general, syllables with initial vowels or initial sonorant consonants had lower random intercepts for *coef-a*. Syllables with initial fricative or aspirated stops or affricates had higher random intercepts for *coef-a*. The results are consistent with reports of segmental effects on  $f_0$ , where the type of consonant affects the  $f_0$  of the following vowel. An initial voiced consonant has the effect of lowering  $f_0$  (Abramson, 1978; Lea, 1973; Löfqvist, McGarr, & Honda, 1984; Ohala, 1973). In Mandarin, a vowel initial syllable may have an initial glottal stop with creaky voice that lowers  $f_0$ , while an aspirated consonant or fricative raises  $f_0$  (Howie, 1976; Shih, 2001; Xu & Xu, 2003; Zee, 1980).

#### 3.4.2. Tone coefficient *b*

In the linear mixed effects model of *coef-b*, TLD and tone were included as fixed effects. The interaction term of TLD and tone was also included, as both a visual inspection of the interaction plots and a model comparison showed an interaction between TLD and tone for *coef-b*. Syllable and speaker were included in the random intercepts models. Repetition was not included, for it explained 0% of the variance in the data. The model explained 67% of the variance in the data. The results are presented in Tables 16 and 17, with speaker effects described in text.

The intercept of the fixed effects was  $-0.025$ . For each increasing level of TLD, *coef-b* increased by 0.0022, though both the intercept and the TLD effect did not reach the significance level of  $p < 0.05$ . Tone 2, tone 3 and tone 4 reduced *coef-b* relative to tone 1. The tone scale for *coef-b* was  $\text{tone1} > \text{tone4} > \text{tone2} > \text{tone3}$ . Tone effects on *coef-b* were significant at  $p < 0.001$  for tone 2 and tone 3 and  $p < 0.01$  for tone 4. The interaction of TLD and tone 3 was significant at the  $p < 0.001$  level. The interaction effects of TLD and other tones were not significant. The random intercepts for speaker were  $-0.059$  for SC (female), 0.087 for SD (female) and  $-0.028$  for SL (male). Table 17 presents the random intercepts adjusting *coef-b* by syllable, showing that syllables with an initial aspirated consonant or fricative tend to have lower random intercepts for *coef-b*, while syllables with an initial vowel or sonorant consonant tend to have higher random intercepts for *coef-b*.

#### 3.4.3. Tone coefficient *c*

A linear mixed effects model was built to predict the value of *coef-c*. Talker-to-listener distance explained very little of the variation in the data (less than 4%). Tone, on the other hand, explained 71.6% of the variance in the data. As in the model of *coef-b*, repetition had no effect and was not included in the random intercepts model. In the final model, TLD and tone were included as fixed effects.

**Table 15**  
Random intercepts adjusting *coef-a* by syllable.

|                  |        |        |        |        |        |
|------------------|--------|--------|--------|--------|--------|
| Syllable         | ai     | wang   | wei    | ya     | cai    |
| Random intercept | -1.203 | -0.602 | -0.459 | -0.408 | -0.374 |
| Syllable         | ma     | yin    | wu     | e      | ju     |
| Random intercept | -0.371 | -0.075 | -0.058 | 0.040  | 0.212  |
| Syllable         | tong   | qiao   | chuan  | guo    | shu    |
| Random intercept | 0.428  | 0.505  | 0.589  | 0.681  | 1.095  |

**Table 16**Summary of the model predicting coef-*b* from fixed effects TLD and tone, along with their interaction, and random effects syllable and speaker.

|                 | AIC<br>–4938                      | BIC<br>–4861                       | logLik<br>2480 | Deviance<br>–4960                    | df.resid<br>7948 |
|-----------------|-----------------------------------|------------------------------------|----------------|--------------------------------------|------------------|
| Random effects: | Groups<br>syll<br>spk<br>Residual | Name<br>(Intercept)<br>(Intercept) |                | Std. dev.<br>0.057<br>0.064<br>0.176 |                  |
| Fixed effects:  | Estimate                          | Std. error                         | <i>t</i> value | <i>p</i> value                       |                  |
| (Intercept)     | –0.0251                           | 0.041                              | –0.61          | 0.54                                 |                  |
| TLD             | 0.0022                            | 0.001                              | 1.78           | 0.076                                |                  |
| tone 2          | –0.2112                           | 0.012                              | –17.62         | <0.001                               |                  |
| tone 3          | –0.4892                           | 0.012                              | –40.79         | <0.001                               |                  |
| tone 4          | –0.0318                           | 0.012                              | –2.65          | <0.01                                |                  |
| TLD:tone 2      | 0.003                             | 0.002                              | 1.67           | 0.094                                |                  |
| TLD:tone 3      | –0.0166                           | 0.002                              | –9.41          | <0.001                               |                  |
| TLD:tone 4      | –0.0001                           | 0.002                              | –0.04          | 0.971                                |                  |

**Table 17**Random intercepts adjusting coef-*b* by syllable.

|                  |        |        |        |        |        |
|------------------|--------|--------|--------|--------|--------|
| Syllable         | qiao   | chuan  | guo    | tong   | shu    |
| Random intercept | –0.106 | –0.084 | –0.062 | –0.046 | –0.042 |
| Syllable         | cai    | e      | ma     | ju     | wang   |
| Random intercept | –0.020 | 0.008  | 0.021  | 0.021  | 0.040  |
| Syllable         | wei    | ya     | ai     | wu     | yin    |
| Random intercept | 0.040  | 0.043  | 0.057  | 0.060  | 0.071  |

**Table 18**Summary of the model predicting coef-*c* from fixed effects TLD and tone and random effects syllable and speaker.

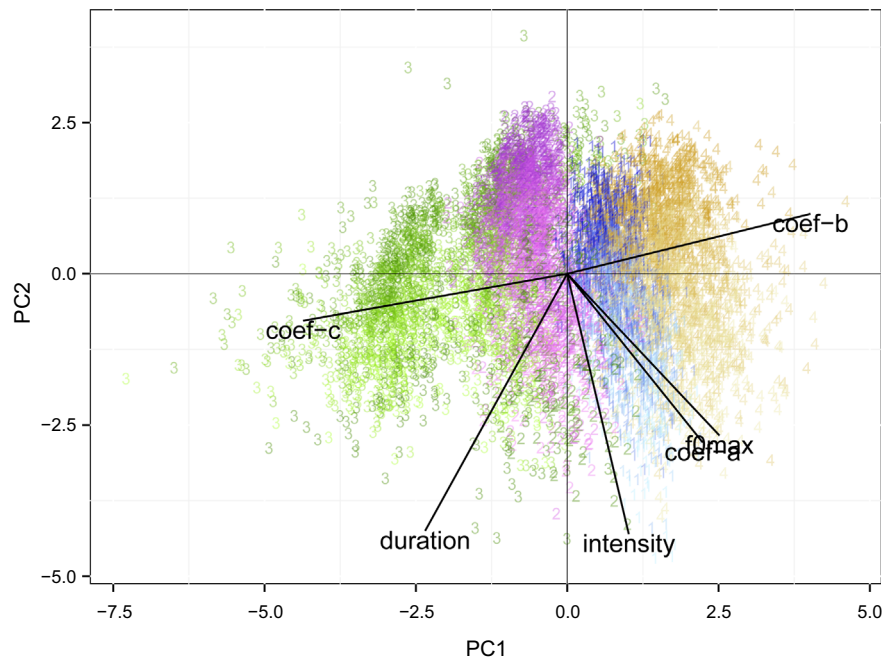
|                 | AIC<br>–67,214                    | BIC<br>–67,158                     | logLik<br>33,615 | deviance<br>–67,230                        | df.resid<br>7951 |
|-----------------|-----------------------------------|------------------------------------|------------------|--------------------------------------------|------------------|
| Random effects: | Groups<br>syll<br>spk<br>Residual | Name<br>(Intercept)<br>(Intercept) |                  | Std. dev.<br>0.00098<br>0.00103<br>0.00353 |                  |
| Fixed effects:  | Estimate                          | Std. error                         | <i>t</i> value   | <i>p</i> value                             |                  |
| (Intercept)     | 0.00009                           | 0.0007                             | 0.14             | 0.89                                       |                  |
| TLD             | 0.00002                           | 0.0000                             | 1.80             | 0.073                                      |                  |
| tone 2          | 0.00545                           | 0.0001                             | 48.74            | <0.0001                                    |                  |
| tone 3          | 0.01107                           | 0.0001                             | 99.05            | <0.0001                                    |                  |
| tone 4          | –0.00509                          | 0.0001                             | –45.56           | <0.0001                                    |                  |

**Table 19**Random intercepts adjusting coef-*c* by syllable.

|                  |           |           |           |           |           |
|------------------|-----------|-----------|-----------|-----------|-----------|
| Syllable         | yin       | ya        | wu        | ai        | wei       |
| Random intercept | –0.001228 | –0.001024 | –0.001023 | –0.000736 | –0.000662 |
| Syllable         | wang      | ma        | e         | ju        | cai       |
| Random intercept | –0.000535 | –0.000447 | –0.000333 | –0.000333 | 0.000465  |
| Syllable         | guo       | shu       | tong      | chuan     | qiao      |
| Random intercept | 0.000754  | 0.000766  | 0.001065  | 0.001586  | 0.001687  |

Syllable and speaker were included in the random intercepts models. The model explained 75.5% of the variation in the data. The results are summarized in Tables 18 and 19.

The intercept of the fixed effects was 0.00009. For each increasing level of TLD, coef-*c* increased by 0.00002. Tone 2 (0.00545) and tone 3 (0.01107) increased coef-*c* relative to tone 1, and tone 4 decreased coef-*c* (–0.00509). The effect of TLD was not significant at the level of  $p < 0.05$ , while the effects of all tones were significant at the level of  $p < 0.0001$ . Tone 3 had the biggest curvature, and Tone 2 and tone 4 had similar magnitudes in curvature but in opposite directions. The random intercepts for speakers were 0.000679 for SC (female), –0.001413 for SD (female) and 0.000734 for SL (male). Table 19 presents the random intercepts adjusting coef-*c* by syllable. Syllables with an initial sonorant consonant or an initial vowel tend to have lower random intercepts for coef-*c*, while syllables with an aspirated initial stop/affricate or an initial fricative tend to have higher random intercepts for coef-*c*.



**Fig. 7.** Principal component analysis of six acoustic attributes: log intensity, duration, maximum  $f_0$  and three tone coefficients. Each data point represents the six acoustic measurements of a sound file projected to PC1 and PC2 after principal component analysis. PC1 is plotted on the x-axis and PC2 on the y-axis. Tone numbers are used as the plotting symbols, with colors added to differentiate the data by tone and shading added to differentiate TLD. The tone colors are: tone 1 in blue, tone 2 in purple, tone 3 in green and tone 4 in yellow. Close TLDs are in darker shades, while farther TLDs are in lighter shades.

### 3.5. Principal component analysis

Correlation studies showed strong multicollinearity among acoustic measurements with TLD and among pairs of acoustic measurements; thus a principal component analysis (PCA) was conducted to find linearly uncorrelated variables to account for the data. Six acoustic measurements discussed earlier were used in the PCA, including three tone coefficients (coef-a, coef-b, coef-c),  $f_0$ max (maximum  $f_0$ ), duration and log intensity. The tone coefficients of an  $f_0$  curve were obtained by the quadratic decomposition of an duration-normalized  $f_0$  curve represented in semitones. The data matrix are scaled using Z-scores so that all variables have zero mean and unit variance.

The first two principal components, PC1 and PC2, explained 68% of the variance in the data, and the first four principal components accounted for 95% of the variance in the data. The results showed that the six acoustic measurements can be reduced to four dimensions without too much loss in explanatory power.

Fig. 7 plots the results of the PCA analysis. Each data point represents the six acoustic measurements of a sound file. The data points are projected to a two-dimensional space, with PC1 as the x-axis and PC2 as the y-axis. Tone numbers are used as the plotting symbols, with colors added to differentiate the data by tone and shading added to differentiate TLD. The tone colors are tone 1 in blue, tone 2 in purple, tone 3 in green and tone 4 in yellow. Close TLDs are in darker shades, while farther TLDs are in lighter shades.

The black lines that extend from the origin indicate the loadings of a factor onto the principal components. The projections of the point at the end of a line on the x- and y-axes indicate the degree of influence the factor has on PC1 and PC2, respectively. Among the six factors, coef-c and coef-b had the biggest absolute loading on PC1, while log intensity and duration had the biggest absolute loading on PC2, indicating their power in explaining the variation in the data.

Eqs. (5) and (6) define the rotation of PC1 and PC2

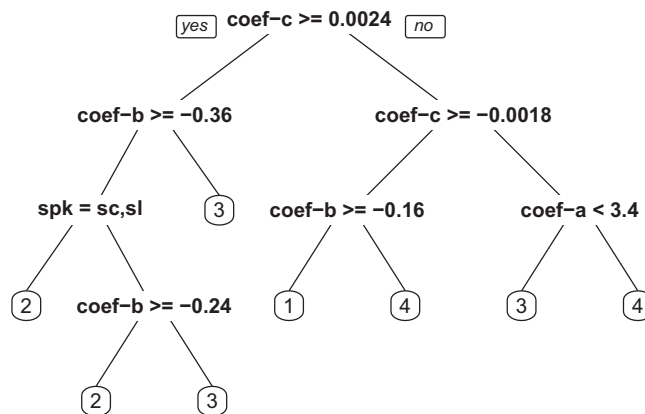
$$PC1 = -0.5989 \cdot \text{coef}_c - 0.3223 \cdot \text{duration} + 0.1396 \cdot \text{intensity} + 0.3072 \cdot \text{coef}_a + 0.3453 \cdot f_0\text{max} + 0.5517 \cdot \text{coef}_b \quad (5)$$

$$PC2 = -0.5906 \cdot \text{intensity} - 0.5834 \cdot \text{duration} - 0.3827 \cdot \text{coef}_a - 0.3669 \cdot f_0\text{max} - 0.1064 \cdot \text{coef}_c + 0.1359 \cdot \text{coef}_b \quad (6)$$

### 3.6. Classification trees

This section investigates the contributions of acoustic measures in the classification of tones using CART, Classification and Regression Tree (Breiman et al., 1984). The R-package *rpart* (Therneau, Atkinson, & Ripley, 2012) was used to perform the classification task.

A CART was built to investigate which of the following factors are most relevant to tone classification: TLD, speaker, syllable, log intensity, duration and tone coefficients coef-a, coef-b and coef-c.



**Fig. 8.** CART: classification of lexical tones using TLD, log intensity, duration, speaker, syllable, and tone coefficients. Only tone coefficients and speaker information are used by CART. The two most effective factors for tone classification are *coef-b* and *coef-c*.

The following classification setting was used: 20 observations were required to split a node, six observations were required in any terminal node, and the classification was done with 10-fold cross-validation.

**Fig. 8** presents the results of the CART. Each node in the tree specifies a factor and a threshold that is used to partition the population into two subgroups. The root node on top represents the entire population. In **Fig. 8**, the factor *coef-c* is used by the root node to make the first cut, dividing the population into two subgroups using the threshold  $\text{coef-c} \geq 0.0024$ . If the condition is met (yes), the data point goes with the left branch, otherwise (no) it goes to the right. The results of the classification are shown in the terminal nodes, with tone numbers enclosed in ovals.

The root node division in **Fig. 8** allocates most data points for tone 2 and tone 3 to the left branch and most data points for tone 1 and tone 4 to the right branch. Following the left branch of the tree, data with  $\text{coef-c} \geq 0.0024$  are tone 2 and tone 3. Then, if  $\text{coef-b} > -0.36$ , it is identified as tone 3. Otherwise, speaker information is needed. For speakers SC and SL,  $\text{coef-b} > -0.36$  and  $\text{coef-c} \geq 0.0024$  are sufficient to identify tone 2. For speaker SD, the threshold of *coef-b* that differentiates between tone 2 and tone 3 is higher than SC and SL: if  $\text{coef-b} \geq -0.24$ , it is identified as tone 2. Otherwise it is identified as tone 3.

There is one pocket of tone 3 data points in the right branch that requires information from  $\text{coef-a} < 3.4$  to differentiate from tone 4.

Tone classification by CART using factors including TLD, log intensity, duration, speaker, syllable and tone coefficients achieved a 93% accuracy rate, with 591 errors out of 7959 cases. All three tone coefficients were used in the classification tree. Additionally, speaker information was used to distinguish tone 2 and tone 3. The other factors did not contribute to tone classification in CART.

#### 4. Discussion of the production experiment

The purpose of the production experiment is to examine changes in acoustic correlates that reflect talker–listener distance (TLD) and their consequences for Mandarin tone production.

Speakers intuitively adjust vocal effort to project their voices over long distances when their listeners are far away and to reduce vocal effort when their listeners are nearby. The changes in vocal effort lead to changes in intensity, duration and *f0*. What is not clear from previous literature is how the *f0* adjustments induced by changing vocal effort affect the production and perception of lexical tones. In this section, we summarize our production experiment's major findings about the acoustic correlates of TLD and their effects on tone, which paved way for the perception study outlined in **Section 5**.

The effects of TLD on acoustics are robust and clear. Log intensity increases monotonically with TLD. This effect is consistent with previous literature (Eriksson & Traünmüller, 2002; Liénard & Benedetto, 1999; Pelegrín-García et al., 2011; Thompson & Webster, 1964; Traünmüller & Eriksson, 2000). Within the range of tested TLDs, the increase in log intensity in response to TLD is linear. Duration and maximum *f0* increase with TLD as well, though there are interaction effects with tones. For every increasing level of TLD (0.51 m in our experimental setting), log intensity increases by 0.09, duration increases by 1.1 centisecond, and maximum *f0* increases by 6.9 Hz. Additionally, there are tone effects and tone and TLD interaction effects that are consistent with the properties of tones.

With regard to log intensity, tone 1 has the highest value and tone 3 has the lowest value. This property is expected given the proportion of high *f0* in a tone. Tone 1 is a sustained high tone, tone 2 and tone 4 have *f0* movements with a portion of the tone in the high *f0* range, and tone 3 is mostly in the low *f0* range.

With regard to duration, tone 3 produced in isolation is generally the longest of four Mandarin tones, and tone 4 tends to be the shortest (Chao, 1968; Feng, 1985; Liu, 1924; Shih & Ao, 1996). The long duration of tone 3 reflects the time it takes to execute consecutive downward and upward *f0* movements. The upward *f0* movement of tone 3 is optional in isolated syllables in some Mandarin-speaking regions, including Taiwan, and is absent in all Mandarin-speaking regions when in non-final positions in continuous speech. When the upward movement is absent, tone 3 is the shortest of the four tones. Thus the long duration of tone 3 in

isolation is a phonetic consequence for  $f_0$  movement, not an intrinsic phonological property of the tone. The different production strategies for tone 3 and their consequences on duration explain the nonsignificant effect of tone 3 in the mixed effects model of duration in Table 7. The shorter duration of tone 4 can also be explained by articulatory phonetics. A speaker's default  $f_0$  setting is closer to the speaker's  $f_0$  floor, with more room to raise pitch than to lower it. A tone 4 is thus terminated when the falling  $f_0$  reaches a speaker's  $f_0$  floor. Table 7 shows that the duration of tone 4 is significantly different from that of tone 1. (Tone 1 was used as the intercept in the fixed effects model.)

The interactions of tone and TLD shown in Table 7 are all significant, and the duration contrast between tones is preserved under different levels of TLD. The duration contrast between tone 1 and tone 3 is enhanced as TLD increases, suggesting that the  $f_0$  movement of tone 3 is exaggerated as TLD increases. Interestingly, the duration contrast between tone 4 and tone 2 is reduced as TLD increases, as shown by the larger coefficient of TLD by tone 4 than TLD by tone 2. Following the model estimation presented in Table 7, the duration difference between tone 1 and tone 3 at TLD level 1 is 5.15 centiseconds. By TLD level 11, the difference almost doubles to 10.05 centiseconds. In contrast, the difference between tone 2 and tone 4 is 7.25 centiseconds at TLD 1, and that difference is reduced to 6.97 centiseconds at TLD 11. This effect is consistent with the phonetic explanation of tone 4 presented above. The increase in  $f_0$  along with TLD provides more  $f_0$  space for the falling gesture of tone 4, thus reducing the shortening effect of a speaker's  $f_0$  floor on tone 4. In summary, TLD has a general effect of increasing duration along with distance, while there is no general effect of enhancing the duration contrast between tones. The interaction of TLD and tone can be explained by the articulatory characteristics of tones.

The maximum  $f_0$  of tones naturally observes their respective phonetics/phonological specifications, with tone 4 having the highest maximum  $f_0$  and tone 3 the lowest. The model-predicted intercept of maximum  $f_0$  in Table 11 is 220 Hz, which is an estimation abstracted from speaker and gender. The TLD effect on maximum  $f_0$  is robust, with an increase of 6.9 Hz for every increased level of TLD. The accumulative effect is considerable. By TLD 11, the maximum  $f_0$  increase is estimated to be 76 Hz, which is more than the differences between tones and is close to the difference between male and female genders at TLD 1. The effect of TLD on the maximum  $f_0$  of each tone is again robust and significant: the maximum  $f_0$  of every tone increases with TLD.

Does TLD also enhance the degree of contrast between the maximum  $f_0$  of tones? The findings of Levow (2006) and Surendran and Levow (2006) suggest that there should be an enhancement effect. This study shows that the spaces between Mandarin tones are more separated under emphatic focus to the extent that lexical tone recognition using machine learning methods is actually easier, even though the tone shapes may look distorted to human eyes. Based on the fixed effects model presented in Table 11, the maximum  $f_0$ s of the four Mandarin tones are 227, 203, 176 and 250 Hz at TLD 1 for tones 1–4, and 297, 278, 221 and 321 Hz at TLD 11 for tones 1–4. The contrasts between tone pairs are enhanced at TLD 11 only if tone 3 is involved. For example, the difference in maximum  $f_0$  between tone 1 and tone 3 is 52 Hz at TLD 1 and increases to 76 Hz at TLD 11. Between tone 3 and tone 4, the difference increases from 74 Hz at TLD 1 to 100 Hz at TLD 11. In contrast, the differences among tone 1, tone 2 and tone 4, though preserved across TLD, are similar or even slightly reduced at TLD 11 in comparison to TLD 1. For example, the difference between tone 1 and tone 2 is 24 Hz at TLD 1 and 19 Hz at TLD 11. The difference between tone 1 and tone 4 is 23 Hz at TLD 1 and 24 Hz at TLD 11. This result shows that the enhancement effect of TLD on contrasts in maximum  $f_0$  is restricted to the contrast between high and low  $f_0$  registers.

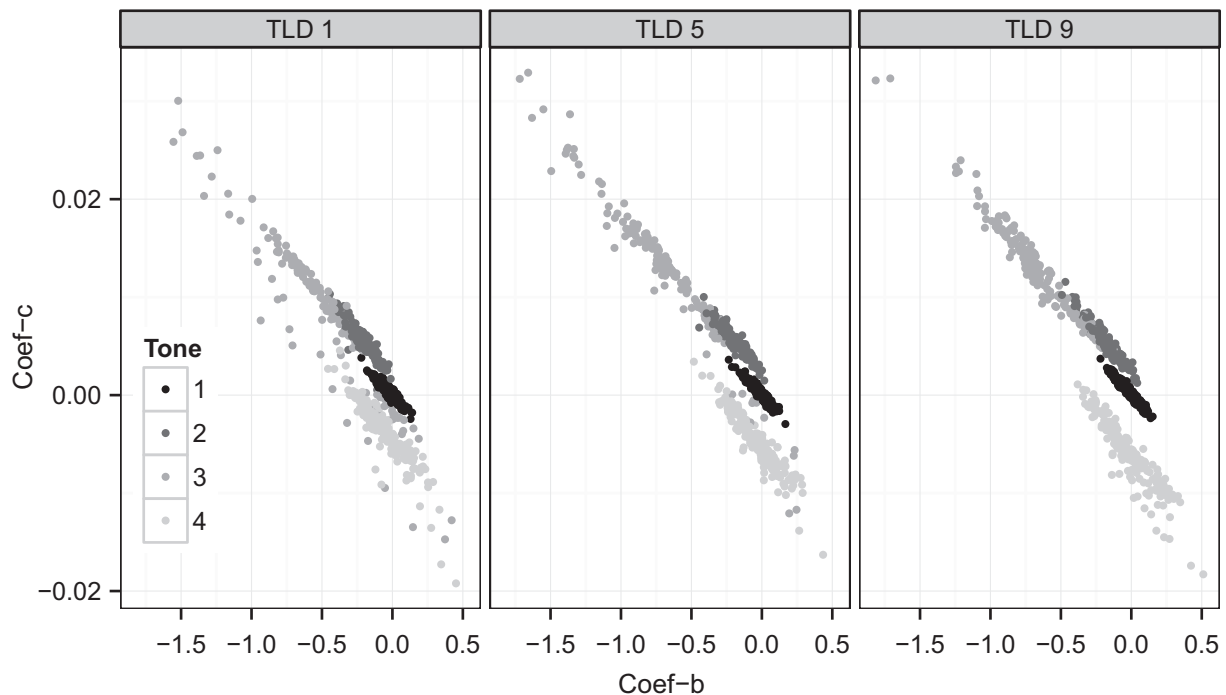
The maximum  $f_0$  is only one point on the  $f_0$  contour. While the increase in maximum  $f_0$  with TLD is a robust effect, one remaining question is how other aspects of tone change with TLD. As explained in Section 2.2, the  $f_0$  contour of a Mandarin syllable can be represented by a part of a parabola using three coefficients of a quadratic function. We scaled all  $f_0$  contours to 50 samples; thus the part of the parabola that is used to represent the tone extends from sample 1 to sample 50. Coefficient  $a$  is the value of the  $y$ -intercept where the parabola crosses the line  $x=0$ . This is one sample that precedes the beginning of the  $f_0$  contour, and is thus a good approximation of the initial  $f_0$  height. This value is different from the maximum  $f_0$  discussed above, which in principle can be any point on the  $f_0$  contour. Coefficient  $b$  determines the position and height of the vertex of the parabola and models the timing and magnitude of the turning point in the tonal contour. Coefficient  $c$  models the curvature of the tonal contour.

The results from mixed effects models presented in Tables 14, 16 and 18 show that the tone effects on tone coefficients are all significant, as expected. Interestingly, the responses from the three tone coefficients to changes in TLD are quite different. Coefficient  $a$  increases with TLD in a manner similar to maximum  $f_0$ . In contrast, TLD effects on tone coefficients  $b$  and  $c$  are weak and not significant.

Parallel to the TLD effect on maximum  $f_0$ , the TLD effect on coef- $a$  is clear and strong, suggesting a common strategy of adjusting  $f_0$  height to account for distance. As TLD increases, coef- $a$  rises accordingly. Following the model displayed in Table 14, tone 4 has the highest coef- $a$  and tone 2 the lowest. The model's prediction shows the same pattern when the means of coef- $a$  are plotted by TLD and tone (Fig. 6). For each increasing level of TLD, coef- $a$  increases by 0.39 semitones up to 4.25 semitones at TLD 11. The magnitude of the increase exceeds the contrast between most tone pairs at TLD 1. If  $f_0$  height is used by L2 listeners to differentiate between tones as reported in Huang and Johnson (2010), we expect that the TLD effects on maximum  $f_0$  and coef- $a$  will cause tone recognition problems.

Table 16 shows that the tone effects on coef- $b$  are strong and significant. There is a slight increase in coef- $b$  along with TLD, but the effect does not reach the significance level of  $p < 0.05$ . There is a significant interaction effect between tone 3 and TLD, as we can see in the interaction plot in Fig. 6. The value of coef- $b$  decreases when TLD increases for tone 3, while the other tones are mostly unaffected by TLD. Similarly, Table 18 shows that the tone effects on coef- $c$  are strong and significant, while the TLD effect again does not reach the significance level of  $p < 0.05$ .





**Fig. 9.** Data from three TLD levels presented in three subplots showing the distribution and the negative correlation of *coef-b* and *coef-c* for the four Mandarin tones. Tones are coded by different shades of gray: tone 1 in black, tone 2 in dark gray, tone 3 in medium gray and tone 4 in light gray. With the exception of some overlapping between tone 2 and tone 3, the properties of the tones are well-described by the negative covariance relationship between *coef-b* and *coef-c*. The lack of variance with regard to TLD is demonstrated by the similarity of the three subplots.

The lack of variance in *coef-b* and *coef-c* under changes in TLD may initially come as a surprise, for the delay of tone targets in Mandarin under emphatic conditions and thus the consequent changes in tone curvature, have been confirmed in studies by different research groups (Shih, 1988; Xu, 2001; Xu et al., 1999). In the current dataset, tone curvature indeed changes with TLD in *f0* contours not normalized by duration. The invariance of *coef-b* and *coef-c* is more evident after duration is normalized. What this means is that there are variations in TLD-related tone curvature that can be explained by duration.

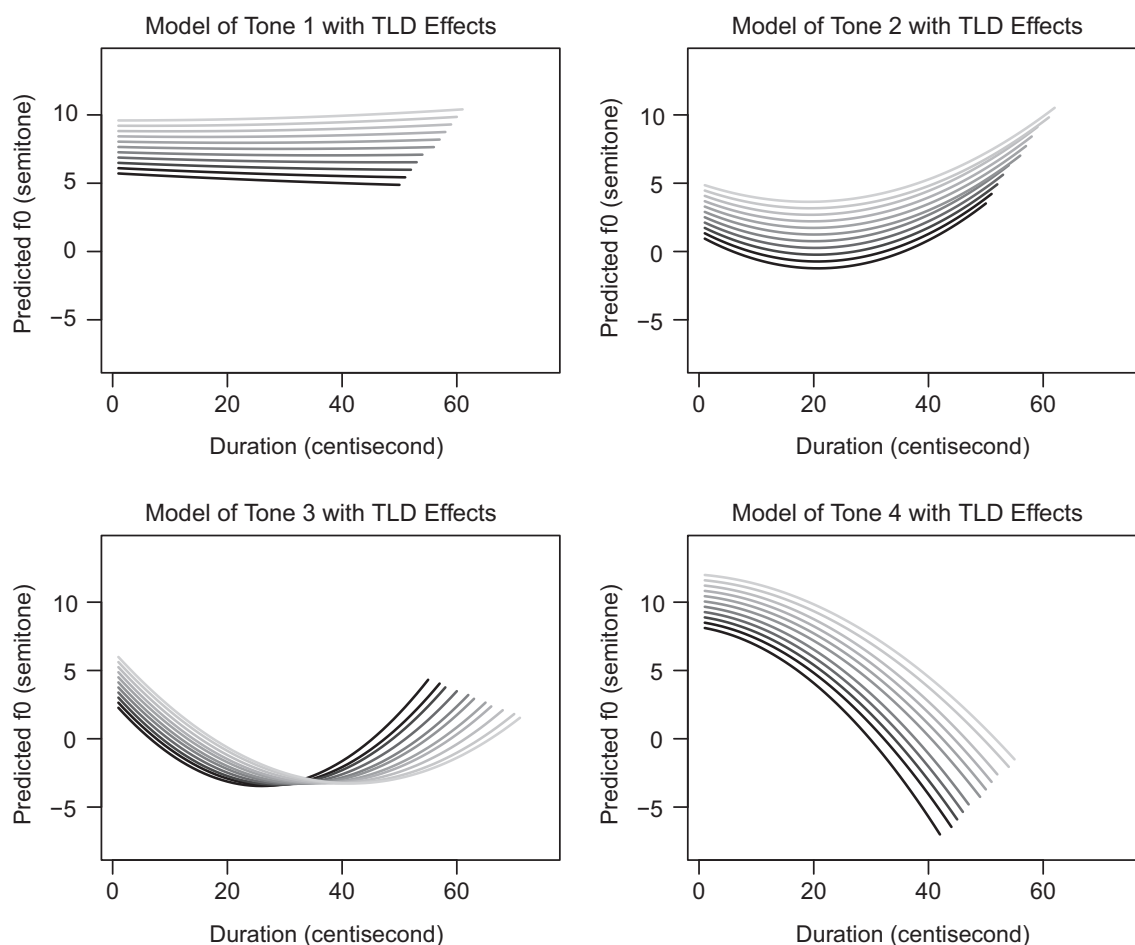
Fig. 9 helps to explain how *coef-b* and *coef-c* define Mandarin tones and how they are invariant under changes in TLD. Three scatter plots present data from TLD 1, TLD 5 and TLD 9 from left to right. In each subplot, *coef-b* is plotted on the x-axis and *coef-c* on the y-axis. Tones are coded in shades of gray: tone 1 in black, tone 2 in dark gray, tone 3 in medium gray and tone 4 in light gray. Each subplot shows the strong collinearity of *coef-b* and *coef-c* and illustrates how they jointly define each Mandarin tone. Populations of different tones are close to one another but mostly separated, and the tone clusters appear to run parallel to each other. Tone 1, in black, forms the tightest cluster, with both *coef-b* and *coef-c* near 0. Tone 2 lies above tone 1, with positive *coef-c*, and tone 4 lies below tone 1, with negative *coef-c*. Tone 3 appears to extend from tone 2, with smaller values for *coef-b* and larger values for *coef-c*. Most tones can already be separated in this two-dimensional space, with the exception of some overlapping between tone 2 and tone 3. The properties of the tones are well-described by the values and the negative covariance of *coef-b* and *coef-c*. The lack of variance with regard to TLD is demonstrated by the similarity of the three subplots. There is no obvious change in the distribution of *coef-b* and *coef-c* for each tone when TLD changes.

The stability of *coef-b* and *coef-c* can be further confirmed with a principal component analysis, and their robustness for tone classification can be confirmed with a CART classification. In the results of PCA presented in Fig. 7, PC1 is the dimension that reflects tonal distinction, and it is primarily explained by *coef-b* and *coef-c*, with some contributions by maximum *f0*, *coef-a* and duration. The values for *coef-b* and *coef-c* move in opposite directions, both with heavy loading on PC1. We see that the four lexical tones form separate groups along this dimension in the following order: tone 4, tone 1, tone 2 and tone 3 (from right to left). Thus, PC1 is dominated by the influence of tone.

The most important direction orthogonal to PC1 is PC2, which reflects TLD. Data points at the top of the figure (in darker shades) are at closer distances, while data points at the bottom of the figure (in lighter shades) are at farther distances. Log intensity, duration, *coef-a* and maximum *f0* (*f0max*), in that order, can mostly explain PC2. The values for *coef-a* and *f0max* are very close to each other, as can be expected because these two factors both reflect *f0* height. They have loadings on both PC1 and PC2, suggesting that they respond to both tone and TLD. Among tone coefficients, *coef-a* is the one with the clearest response to TLD.

The results of the CART further confirm the relative importance of *coef-b* and *coef-c* among acoustic attributes for tone classification. In Fig. 8, *coef-c* and *coef-b* are the most effective factors that can be used to separate lexical tones. A high, positive value for *coef-c* predicts tone 2 or tone 3, a small magnitude predicts tone 1, and a large negative value predicts tone 4. Coefficient *b* is useful for separating tone 2 from tone 3 and tone 1 from tone 4. Coefficient *a* is only used to separate a small cluster of tone 3 from tone 4, where the third tones were produced with a stronger downward than upward pitch movement. Such shapes may be fitted with





**Fig. 10.** Simulation of TLD effects on tone shape and duration. Generated  $f_0$  curves for different TLD distances are shaded from dark to light gray, representing distances from nearest to farthest.

a negative  $\text{coef-c}$  to depict a parabola with a downward turn. This shape is similar to that of tone 4; thus proper tone classification requires information from  $\text{coef-a}$  to separate a high-falling tone 4 from a low-falling tone 3.

Fig. 10 integrates model predictions to simulate TLD effects on tones. The fixed-effects coefficients from Tables 14, 16 and 18 are used to reconstruct the tone shapes. The tone shapes generated from 50 samples are mapped to the duration predictions from Table 7. The results are abstract toneshapes with TLD effects devoid of speaker, syllable and repetition effects, which can be added by applying the model-predicted random effects. The four tones are plotted separately, with tone shapes predicted for TLD level 1 drawn in dark gray and farther TLD distances in lighter shades of gray.

The production experiment shows clear effects of TLD on log intensity, duration, maximum  $f_0$  and  $\text{coef-a}$ . All these acoustic attributes increase in response to increases in TLD. The contrast between the duration and maximum  $f_0$  of tone 3 and those of other tones is enhanced by increasing TLD, though this enhancement effect is restricted to the contrast between high and low  $f_0$  registers, and is not observed in the contrast between other tone pairs.

Two tone coefficients that control tone shapes,  $\text{coef-b}$  and  $\text{coef-c}$ , remain invariant under changes in TLD. The lack of TLD effects on  $\text{coef-b}$  and  $\text{coef-c}$  stands in sharp contrast to the robust effects of TLD on log intensity, duration, maximum  $f_0$  and  $\text{coef-a}$ . This finding has strong implications for tone identification. We interpret the stability of  $\text{coef-b}$  and  $\text{coef-c}$  as the shared knowledge of tonal contrasts between Mandarin speakers and listeners, in that speakers make an effort to maintain these invariant tone features for successful speech communication.

## 5. The perception experiment

The goal of the perception experiment were to investigate whether TLD effects have an impact on tone perception. Perception data from both native speakers of Mandarin (L1) and Mandarin learners (L2) was collected. Some concepts from item analysis techniques were adopted to examine the difficulty level of a tone stimulus and determine whether a tone stimulus can distinguish L2 listeners in the high performance group from those in the low performance group. We then investigated whether L2 tone identification errors could be explained by the acoustic attributes of the stimuli.

### 5.1. Subjects

Two native Mandarin listeners and 52 L2 Mandarin learners participated in the tone perception task. The two native listeners were teachers of Mandarin language classes at the University of Illinois at Urbana-Champaign.

The 52 L2 listeners were students at the University of Illinois at Urbana-Champaign, though none had been taught by the native speakers or native listeners. Students enrolled in Chinese 202, the second semester of Elementary Chinese, were recruited to participate in the L2 tone perception experiment. Participants who completed the experiment received extra credit in the course. At the time of the experiment, the L2 learners had taken Mandarin Chinese for eight months, with about 125 h of classroom instruction.

### 5.2. Materials

The same materials recorded for the production study described in Section 2 were used for L1 listeners. Twenty-seven files with recording errors and speech errors were deleted from the database by the authors. The resulting tone perception database for L1 listeners consisted of 7959 files.

Native listeners identified all sound files by listening to them in a random order and identifying the tones. Native listener tone error analysis was performed before the L2 perception study. Only files that were correctly identified by both L1 listeners were included in the pool of stimuli chosen for the L2 listeners.

Second-language listeners identified 144 sound files. Only odd-number distance levels were used. Six syllables were randomly chosen from the inventory of 15 syllables in the production study. The selected syllables were: *ai*, *cai*, *ju*, *ma*, *qiao*, and *wang*. The selection of the 144 files followed a randomized block design, where equal numbers of stimuli were chosen for each tone, speaker, syllable, and TLD level, with an additional requirement that they cover every syllable in each combination of tone and TLD. The specific files in each block were then chosen randomly. There were four tones, three speakers, six syllables and six TLD levels; thus there were 36 files for each tone, 48 files for each speaker, 24 files for each syllable, 24 files for each TLD level and six files for the six syllables in each combination of tone and TLD.

### 5.3. Procedure

The informed consent process and the goals and procedures of the experiment were explained to students in the classroom, and UIUC Institutional Review Board (IRB) consent forms were handed out. All students were assigned a password and allowed to register with a web-based system to participate in the tone identification test. The responses of 52 consenting subjects were kept for data analysis.

The tone identification task was administered through a web interface with MySQL and PHP scripts to collect the responses. Speech files were presented one by one. Subjects listened to a sound file first, then clicked the perceived Mandarin tone. Their answers were logged for analysis. This procedure has its limitations. For instance, we were not able to collect reaction time data, control speaker quality and settings, and there was some data loss due to internet problems cutting short four of the tests.

For the L1 listeners, speech files were divided into 12 blocks, with a randomized order within each block. Each file was played only once, and listeners were encouraged to choose the tones they perceived as quickly as possible. They were allowed to take breaks as often as they wanted.

For the L2 listeners, the 144 speech files were presented in one block in a randomized order. The listeners were allowed to listen to a speech file repeatedly, though they could not go back to a previous file.

## 6. Analysis of the perception study

This section analyzes L1 and L2 tone perception data. One question this analysis aims to answer is whether TLD-induced tone variations lead to tone perception problems. In the following, we present tone confusion matrices for L1 and L2 listeners, descriptive statistics, and results from mixed effects models predicting item difficulty and item discrimination.

**Table 20**

Tone confusion matrix for L1 listeners. The row names represent actual tones, and the column names represent identified tones.

|        | Tone 1         | Tone 2         | Tone 3         | Tone 4         | Total |
|--------|----------------|----------------|----------------|----------------|-------|
| Tone 1 | 4000<br>99.85% | 1<br>0.03%     | 1<br>0.03%     | 4<br>0.10%     | 4006  |
| Tone 2 | 4<br>0.10%     | 3921<br>98.84% | 39<br>0.98%    | 3<br>0.08%     | 3967  |
| Tone 3 | 1<br>0.03%     | 156<br>3.94%   | 3803<br>96.04% |                | 3960  |
| Tone 4 | 1<br>0.03%     | 2<br>0.05%     | 4<br>0.10%     | 3963<br>99.82% | 3970  |

**Table 21**

Tone confusion matrix for L2 listeners. The rows represent actual tones and the columns represent identified tones.

|        | Tone 1         | Tone 2         | Tone 3         | Tone 4         | Total |
|--------|----------------|----------------|----------------|----------------|-------|
| Tone 1 | 1539<br>87.39% | 128<br>7.27%   | 22<br>1.25%    | 72<br>4.09%    | 1761  |
| Tone 2 | 172<br>9.40%   | 1264<br>69.07% | 336<br>18.36%  | 58<br>3.17%    | 1830  |
| Tone 3 | 50<br>2.66%    | 281<br>14.92%  | 1496<br>79.45% | 56<br>2.97%    | 1883  |
| Tone 4 | 119<br>6.68%   | 96<br>5.39%    | 29<br>1.63%    | 1538<br>86.31% | 1782  |

### 6.1. Tone confusion matrices

Native listeners' identification of monosyllabic tones was robust even under suboptimal conditions, with an accuracy rate of 98.64%. Table 20 presents the confusion matrix for L1 tone identification. Data from both listeners are combined. The row names represent actual tones, and the column names represent identified tones. Each cell gives the count for the cell followed by the percentage (the count divided by the row sum). The last column provides the row sum, or the token count for each tone. There were 216 tone identification errors, including six tone 1 errors, 46 tone 2 errors, 157 tone 3 errors and seven tone 4 errors. The majority of these errors were caused by confusion between tone 2 and tone 3.

There was no sign that TLD had an impact on L1 tone perception. The distribution of L1 errors was not correlated with TLD ( $R = 0.14, t = 0.4288, df = 9, p = 0.6782$ ).

Table 21 presents the tone confusion matrix for L2 learners. The format follows that of Table 20. The row names represent actual tones, and the column names represent identified tones. Errors of all types were observed. As with L1 listeners, most of the errors were caused by confusion of tone 2 and tone 3. Collectively, there were 1419 L2 errors made by 52 subjects, with a tone identification accuracy rate of 80.44%. The number of errors per subject ranged from 2 to 59, and the average was 27 errors out of 144 items.

Second-language learners made more tone identification errors even though the L2 task was easier than the L1 task. All files presented to L2 listeners had been correctly identified by L1 listeners; thus the most confusing files as well as possible production errors were excluded. The scope of the tasks was different as well. Native listeners identified 7959 files each, while L2 listeners identified 144 files each. The L1 task was longer and thus more prone to the effects of fatigue.

### 6.2. Item analysis of L2 tone perception errors

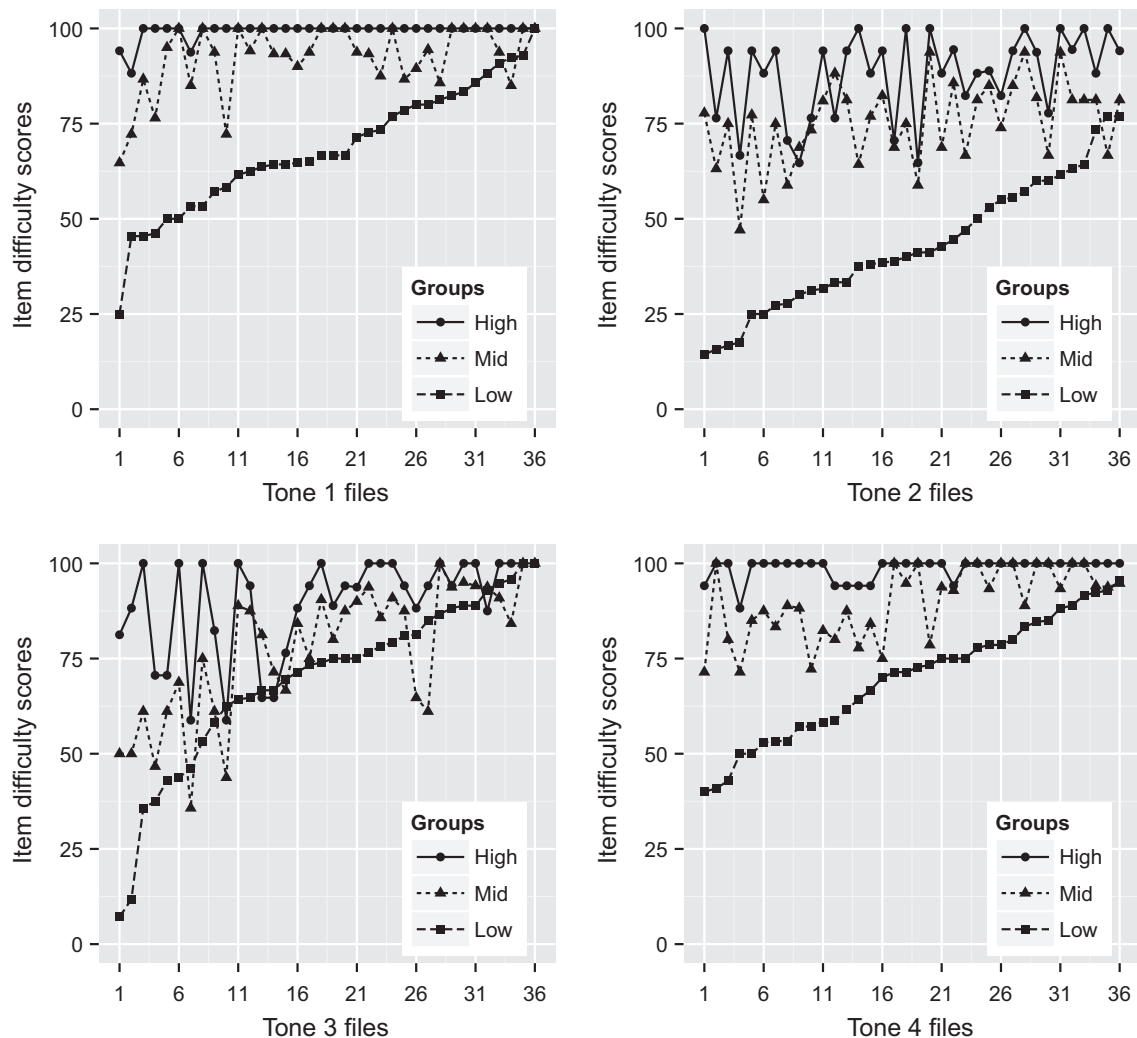
Item analysis is commonly used in the testing community to analyze how students respond to each item of a test in order to evaluate the validity of both individual test items and the test as a whole (van der Linden & Hambleton, 1996). Test items that can successfully differentiate subjects by the characteristics being tested are considered good items that strengthen the validity of the test. On the other hand, if there is an item that students with high scores tend to answer incorrectly but students with low scores tend to answer correctly, then there may be a subtle ambiguity to the item that punishes students who think deeply. In traditional item analysis, one of the goals is to identify such problematic items and eliminate them from future tests (Wood, 1960). For the current project, we assumed that tone identification errors made by L1 listeners were sufficient to exclude problematic files. We used the concept of item analysis for a different purpose. Our goal was to locate speech files that may have contained acoustic properties of interest to the study of L2 acquisition of tones, including stimuli that are difficult for L2 learners as well as stimuli that can differentiate between learners at different learning stages.

Two concepts from item analysis were used in the analysis of L2 perception errors, though we did not use the exact formula of item analysis in that we used scores representing the percentage correct rather than scale item difficulty scores to 0–1 and item discrimination scores to –1 and 1.

1. *Item difficulty score*: The percentage of participants who answer an item correctly. High scores indicate easy items while low scores indicate difficult items.
2. *Item discrimination score*: Participants are divided into high, mid and low groups based on their test scores. The item discrimination score of an item is the difference between the item difficulty scores of the high and low groups for that item. In the context of this study, a high discrimination score for a stimulus is suggestive of the presence of acoustic properties in the sound file that can differentiate participants in the high group from those in the low group.

The 52 L2 subjects were divided into three performance groups based on their total tone identification scores: the high, mid and low groups. The high and low groups each consisted of 17 subjects, while the mid group had 18.

Fig. 11 presents the item difficulty scores for the stimuli used in the L2 perception study. The 144 files are separated by tone into four subplots with 36 files each. Each file is represented by three scores, which are the percentage of correct answers given by the high, mid and low groups. A score of 0 means that no one in the group identified the tone of the stimulus correctly, and a score of 100 means that everyone in the group answered correctly.



**Fig. 11.** Item difficulty scores for L2 tone perception stimuli. Stimuli are separated by tone into four subplots with 36 stimuli per plot. Item difficulty scores for high, mid and low groups are plotted by the solid line with circles, dotted line with triangles and dashed line with squares, respectively. Item discrimination scores can be estimated visually from the difference between the high and low groups.

**Table 22**

Item difficulty scores from the low group by TLD and tone.

|        | TLD 1 | TLD 3 | TLD 5 | TLD 7 | TLD 9 | TLD 11 |
|--------|-------|-------|-------|-------|-------|--------|
| Tone 1 | 62    | 73    | 69    | 71    | 70    | 64     |
| Tone 2 | 39    | 43    | 38    | 37    | 50    | 51     |
| Tone 3 | 48    | 58    | 71    | 75    | 78    | 85     |
| Tone 4 | 71    | 80    | 64    | 71    | 72    | 60     |

In each subplot, files are plotted on the x-axis and are ordered by the scores of the low group. The item difficulty scores from the high, mid, and low groups are plotted by the solid line with circles, dotted line with triangles and dash line with squares, respectively. There are many perfect scores from the high group in the tone 1 plot on the top left and the tone 4 plot on the bottom right. The lowest score is 7.14 in the tone 3 plot on the bottom left, which means that only 7% of the low group listeners answered that stimulus correctly.

The item discrimination score for a stimulus is calculated by subtracting the item difficulty score of the low group from the score of the high group. The first two files in the tone 2 plot have the two highest item discrimination scores in the L2 perception database: 85.7 and 77.5. The first file in the tone 3 plot comes in third at 76.5. In the tone 3 plot, there are files with item discrimination scores of zero or in the negative, where the low group performed equal to or better than the high group.

The high-performance group in general performed quite well, with a ceiling effect for tone 1 and tone 4. Their answers were not informative for error analysis, so we concentrated on the low group first. Table 22 presents the item difficulty scores for the low group separated by tone and TLD. There appears to be an interaction of tone and TLD with item difficulty, as shown by the means. Tone 1 seems less affected by TLD, though the two extremes, TLD 1 and TLD 11, produced lower scores. Tone 2 and tone 3 identification shows improvement as TLD increases, while tone 4 identification seems to degrade.

Pearson's product-moment correlation coefficient was used to check whether there were TLD effects on tone perception. The correlation ( $r$ ) scores between low group item difficulty and TLD were 0.02, 0.24, 0.54, and  $-0.23$  for tone 1, tone 2, tone 3 and tone 4, respectively. There was a trend that tone 3 identification became easier as TLD increased. Fig. 12 shows a moderate correlation between item difficulty and TLD for tone 3 in the low group, where item difficulty is plotted as a function of TLD. The regression line  $y = 48.31 + 3.46 \cdot TLD$  ( $R^2 = 0.29$ ), plotted as a black solid line, shows a positive slope, suggesting that each increasing level of TLD produces an improvement of 3.46% in the identification of tone 3 by L2 listeners in the low group.

### 6.3. Mixed effects models predicting item difficulty scores

A mixed effects regression model was built to investigate which factors help to explain L2 tone perception. Item difficulty scores for the 144 files for 52 L2 subjects were used as the dependent variable, and multiple models were compared to test the contributions of different fixed and random effects following the procedure described in Section 3.1.

Syllable and repetition had no significant effects, so the random-intercept models included only the speaker term. Models built with tone and tone coefficients in the fixed effects were significantly better than models without them. Adding log intensity or maximum  $f_0$  led to no significant improvement. Duration had a marginal effect, though its inclusion was not warranted, for the improvement did not outweigh the complexity it added to the model. Talker-to-listener distance had no significant effect because of the presence of an interaction between TLD and tone. When the interaction of tone and TLD was added, the effect was significant. The final model included TLD, tone, the interaction of TLD and tone, and tone coefficients in the fixed effects model. Speaker was included in the random effects model. The number of observations is 144. Groups include speaker (3 levels). The model explained 49% of the variance in the data. The results are presented in Table 23.

The item difficulty scores used in this model ranged from 0 to 100. A high score implies an easy file, while a low score implies a file difficult for L2 listeners. Data from all L2 listeners were used. The intercept of the fixed effects was predicted to be 88. Tone 2 and tone 3 had significant negative effects, reducing the scores by 21 and 27, respectively. The interaction between tone 3 and TLD was significant at the  $p < 0.01$  level. Tone 3 identification improved with an increase in TLD. For every increased level of TLD, L2 listeners' tone 3 perception improved by 1.8%. At TLD level 11, the recognition rate was close to 20% better than the rate at TLD level 1.

The variable speaker in the random effects model explained 6% of the variance in the data. The coefficients were  $-2.27$ ,  $-0.94$  and  $3.21$  for speakers SC, SD and SL, respectively. Data analysis showed that SC produced more third tones without a rising contour than other speakers at close TLDs, which led some L2 listeners to identify tone 3 as tone 4.

### 6.4. Mixed effects models predicting item discrimination scores

We followed the same procedure to build models to predict the item discrimination scores for the tone files, and to understand what acoustic attributes may be used to distinguish L2 learners in the high-performing group from those in the low-performing group. Item discrimination scores for the 144 stimuli, calculated as the percentage of correct answers given by the high group minus the percentage of correct answers given by the low group, were used as the dependent variable.

As in the item difficulty model, the effects of TLD levels, log intensity, duration and maximum  $f_0$  were not significant and were not included in the fixed effects model. Syllable was not significant and was not included in the random effects model. Talker-to-listener distance, tone, TLD and tone interaction as well as tone coefficients all significantly improved the model when tested individually, though TLD as well as the interaction term of TLD and tone lost explanatory power in the presence of tone coefficients. The final

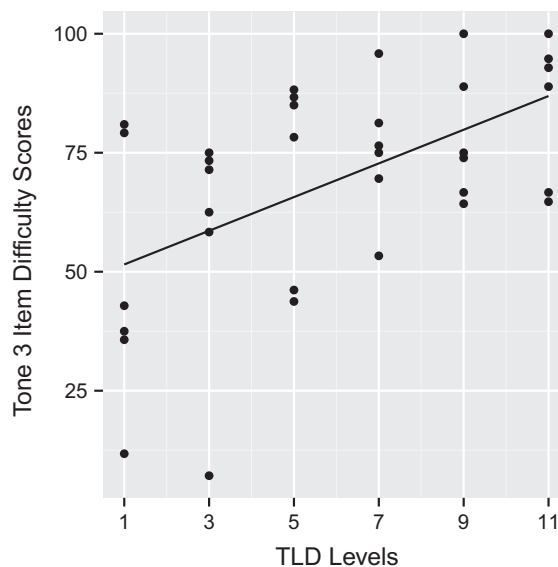


Fig. 12. Item difficulty scores for the low group's performance on tone 3 identification are plotted as a function of TLD. The regression line is plotted by a solid black line. There is a moderate positive correlation showing that tone 3 identification improves with TLD.



**Table 23**

Summary of the model predicting item difficulty scores of the stimuli.

|                 | AIC                       | BIC                 | logLik         | Deviance                  | df. resid |
|-----------------|---------------------------|---------------------|----------------|---------------------------|-----------|
|                 | 1082                      | 1120                | –528           | 1056                      | 131       |
| Random effects: | Groups<br>spk<br>Residual | Name<br>(Intercept) |                | Std. dev.<br>2.62<br>9.30 |           |
| Fixed effects:  | Estimate                  | Std. error          | <i>t</i> value | <i>p</i> value            |           |
| (Intercept)     | 88.12                     | 5.12                | 17.20          | <0.001                    |           |
| tone 2          | –21.41                    | 6.12                | –3.50          | <0.001                    |           |
| tone 3          | –26.83                    | 7.15                | –3.75          | <0.001                    |           |
| tone 4          | 4.88                      | 5.56                | 0.88           | 0.38                      |           |
| TLD             | 0.36                      | 0.56                | 0.64           | 0.52                      |           |
| coef- <i>a</i>  | –0.40                     | 0.77                | –0.52          | 0.60                      |           |
| coef- <i>b</i>  | –7.72                     | 21.39               | –0.36          | 0.72                      |           |
| coef- <i>c</i>  | 186.66                    | 899.51              | 0.21           | 0.84                      |           |
| tone 2:TLD      | –0.19                     | 0.64                | –0.30          | 0.77                      |           |
| tone 3:TLD      | 1.80                      | 0.66                | 2.72           | <0.01                     |           |
| tone 4:TLD      | –0.65                     | 0.67                | –0.98          | 0.33                      |           |

**Table 24**

Summary of the model predicting the item discrimination capability of sound files.

|                 | AIC                       | BIC                 | logLik         | Deviance                   | df. resid |
|-----------------|---------------------------|---------------------|----------------|----------------------------|-----------|
|                 | 1224                      | 1250                | –603           | 1206                       | 135       |
| Random effects: | Groups<br>spk<br>Residual | Name<br>(Intercept) |                | Std. dev.<br>3.05<br>15.74 |           |
| Fixed effects:  | Estimate                  | Std. error          | <i>t</i> value | <i>p</i> value             |           |
| (Intercept)     | 42.56                     | 6.55                | 6.50           | <0.001                     |           |
| tone 2          | 15.96                     | 5.75                | 2.78           | <0.01                      |           |
| tone 3          | –15.53                    | 7.81                | –1.99          | <0.05                      |           |
| tone 4          | –24.43                    | 7.92                | –3.08          | <0.01                      |           |
| coef- <i>a</i>  | –1.55                     | 0.77                | –2.00          | <0.05                      |           |
| coef- <i>b</i>  | –87.31                    | 29.95               | –2.92          | <0.01                      |           |
| coef- <i>c</i>  | –4601.06                  | 1330.55             | –3.46          | <0.001                     |           |

model predicts the item discrimination scores of sound files using tone and tone coefficients as the fixed effects and speaker as the random effect. The number of observations is 144. Groups include speaker (3 levels). The model explained 34% of the variance in the data. A summary of the results is presented in Table 24.

All fixed effects were significant at the  $p < 0.05$  level. Furthermore, tone 2, tone 4 and coef-*b* were significant at the  $p < 0.01$  level, and coef-*c* was significant at the  $p < 0.001$  level. The model started with a prediction of an item discrimination score of 43. Tone 2 was effective at increasing item discrimination scores, while tone 3 and tone 4 had a reducing effect. All tone coefficients had a significant impact on item discrimination. The negative values suggested that item discrimination was worse with larger values for tone coefficients.

The variable speaker in the random effects model explained 5% of the variance in the data. The coefficients were 3.39, –1.97 and –1.42 for speakers SC, SD and SL, respectively. Some of the most difficult sound files were useful for the purpose of item discrimination.

## 7. General discussion

The results of the production study show that some acoustic attributes vary with TLD and some remain invariant. The acoustic attributes that change with TLD include log intensity, duration, maximum *f0* and coef-*a*, which captures the initial *f0* of a tone. Two of the acoustic attributes invariant under changes in TLD include coef-*b* and coef-*c*, the two tone coefficients that capture tone shapes.

Native listeners have excellent tone perception when it comes to speech impacted by TLD. They are able to decode the systematic, monotonic changes in acoustics that reflect TLD and use that information, especially the lengthening of duration, to normalize *f0* contours and obtain invariance features to process tone information. They make very few errors in tone identification, with an accuracy rate of 98.6%. Both the error rate and error types of native listeners are consistent with reports in the previous literature (Gottfried & Suiter, 1997; Howie, 1976; Surendran et al., 2005). The tone accuracy rate of 98.6% for TLD-impacted speech

is nearly identical to native listeners' tone recognition rate of 98.96% under ideal circumstances using intact syllables (Gottfried & Suiter, 1997). Native speakers' error distribution is not correlated with TLD, showing that their tone recognition ability is not affected by changes brought on by TLD.

As expected, the performance of L2 listeners is weaker than that of native listeners. The confusion matrix, error count and mixed effects model predicting item difficulty (Table 23) all show that distinguishing between tone 2 and tone 3 is difficult for L2 listeners. The pattern is similar in direction to that of native listeners, with amplifications in magnitude, and the results are consistent with the previous literature (Kiriloff, 1969; Wang et al., 1999). The difficulty levels of identifying tone 2 and identifying tone 3 are both significant at the level of  $<0.001$ .

Table 23 shows another significant result. There is an interaction between TLD and L2 tone perception, indicating some stages at which L2 listeners have not yet acquired the ability to fully separate variations caused by TLD from variations due to lexical tone classification. The interaction is between tone 3 and TLD. The direction of the effect suggests that tone 3 stimuli recorded at close TLDs are not clear enough for some L2 listeners, while those recorded at far TLDs are clearer. The model predicting item discrimination scores presented in Table 24 provides information about the factors that separate the low and high groups. Tones and tone coefficients all have significant effects. Therefore, we must pay attention to the tone coefficients of files within each tone category that are particularly difficult for L2 listeners in the low group.

Fig. 13 shows the effects of *coef-b* and *coef-c* on the difficulty of identifying tone 1. Coefficient *c* is plotted on the x-axis and *coef-b* on the y-axis. The item difficulty scores for the low group are used as the plotting symbols, with lower scores indicating more difficult files. There are four tone 1 files with low-group item difficulty scores below 50, which are circled in the plot. One file each was produced at TLDs 1 and 3, and two were produced at TLD 11. They all have negative *coef-b* and positive *coef-c* values, with extreme values for tone 1 in the perception study.

The ideal tone 1 shape is level, which implies no curvature and a *coef-c* value close to 0. A larger, positive *coef-c* suggests that the pitch curve becomes increasingly more like that of tone 2. The proportion of fall and rise changes with the magnitude of *coef-b*. With a large negative *coef-b*, the vertex of the parabola can move out of the range of the time-normalized *f0* samples; the described pitch would then be just falling. Many L2 listeners in the low group and some in the mid group identified these files as either tone 2 or tone 4, possibly because they heard a slight pitch movement but could not determine the direction.

There are two data points in the lower right corner of Fig. 13 which have extreme values of *coef-b* and *coef-c* but a relative good item difficulty score of 64. An inspection of the *f0* contours reveals that, in each file, the change in *f0* slope is restricted to the first 150 ms of the syllable, while the rest of the *f0* contour remains flat. It is possible that this pattern is helpful for the identification of tone 1.

There are 23 tone 2 files with item difficulty scores less than 50 for the low group. Considering that there are only 36 tone 2 files in the perception experiment, the error rate is quite high, and the result is consistent with previous studies showing that the highest error rate is for Mandarin tone 2 (Kiriloff, 1969; Sun, 1998). There are signs suggesting that L2 subjects in the low group are actually not able to differentiate tone 2 from tone 3. Results from the literature and from the native listeners as well as the mid and high group L2 listeners who participated in this study all show that tone 2 and tone 3 is the most difficult pairing to differentiate. The low group's ability to identify tone 3 (bottom left in Fig. 11) is thus surprisingly good, with many cases in which they outperform the mid and high groups. Analysis of the response patterns to tone 2 by the low group shows that their answers are not correlated with acoustic measures. One likely explanation is that the low group cannot distinguish tone 2 from tone 3 reliably but tends to choose tone 3 when presented with either a tone 2 or tone 3 stimulus. The tendency to choose tone 3 artificially boosts their performance on the identification of tone 3, with a trade-off in their performance on identifying tone 2.

Tone 2 identification by the mid and high groups was considerably better. There are signs that listeners in the mid group were using information from the tone coefficients to identify tone 2. Their tone 2 scores show moderate correlation with all three coefficients, though none reached the significance level of  $p < 0.05$ . It takes the more advanced skills of listeners in the high group to demonstrate a sensitivity to acoustic information in the identification of tone 2. There are significant correlations between the high

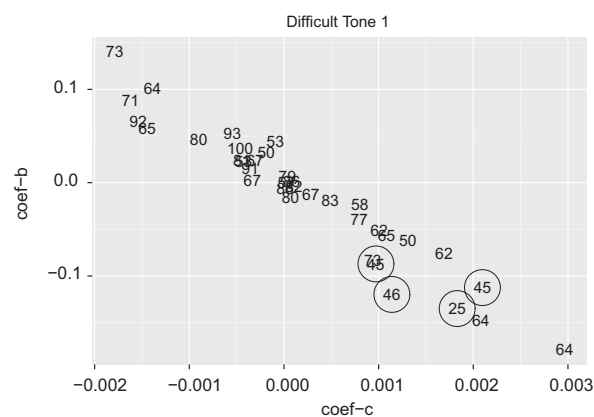
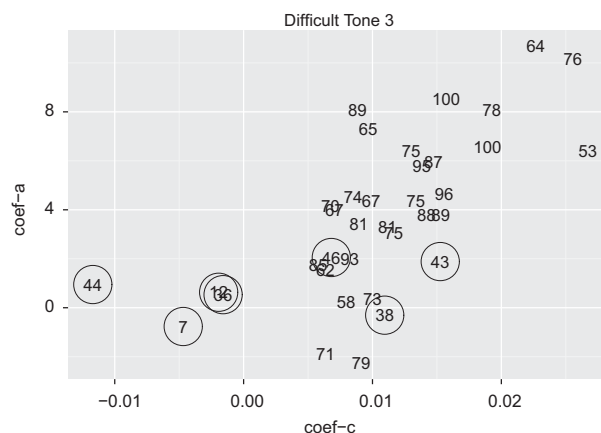
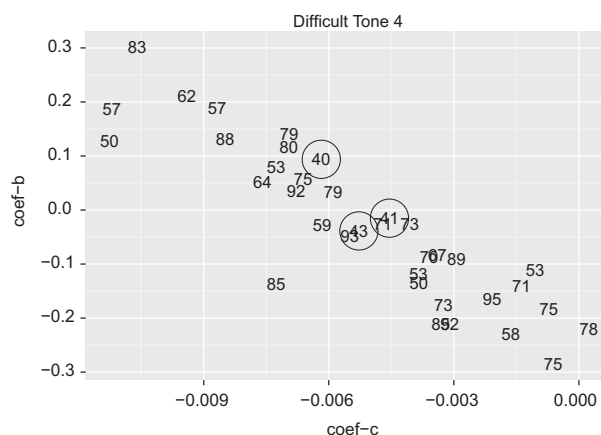


Fig. 13. Difficult tone 1 stimuli. All tone 1 stimuli are plotted with *coef-c* on the x-axis and *coef-b* on the y-axis. The item difficulty scores for the low group are used as the plotting symbols, where the scores range from 0 to 100 with low scores indicating more difficult files. The four files with the worst scores are circled. They have negative *coef-b* and positive *coef-c*, with extreme values for tone 1.



**Fig. 14.** Difficult tone 3 stimuli. All tone 3 stimuli are plotted with coef-c on the x-axis and coef-a on the y-axis. The item difficulty scores for the low group are used as the plotting symbols. The seven worst files are circled. Files with negative coef-c lead to a high error rate.



**Fig. 15.** Difficult tone 4 stimuli. All tone 4 stimuli are plotted with coef-c on the x-axis and coef-b on the y-axis. The item difficulty scores for the low group are used as the plotting symbols. The three most difficult tone 4 files are circled. Tone 4 stimuli with low curvature and shallow slope tend to have higher error rates.

group item difficulty scores and coef-b ( $r=0.426$ ,  $t=2.7493$ ,  $df=34$ ,  $p=0.009$ ) and coef-c ( $r=-0.417$ ,  $t=-2.675$ ,  $df=34$ ,  $p=0.011$ ). The correlation with coef-a has not reached the significant level ( $r=-0.307$ ,  $t=-1.8834$ ,  $df=34$ ,  $p=0.068$ ). The different response patterns to tone 2 by the low, mid and high groups show advances in the ability to process tone information, particularly with attention to the subtle nuances in coef-b and coef-c that differentiate tone 2 from tone 3.

As we have already seen in Fig. 11, the performance of the low group on the identification of tone 3 is surprisingly good. There were only seven files with scores below 50 in contrast to 23 for tone 2. Fig. 14 plots all tone 3 files with coef-c on the x-axis and coef-a on the y-axis. Item difficulty scores for the low group are used as plotting symbols. The seven worst files are circled. All seven were produced at close TLDs within the first five levels, with four produced at TLD level 1.

Four of these seven files are in the lower left corner of the plot, with negative coef-c and a relatively low value for coef-a. They also have relatively large coef-b values (not shown in the plot). These properties lead to a high error rate for L2 listeners in the low and mid groups, who are equally likely to report hearing tone 4 or tone 2. The three circled files at the center of the plot have a higher propensity to be incorrectly identified as tone 2 by L2 listeners in all three performance groups. It seems that tone 3 files are difficult for L2 listeners because the first part of the tone resembles tone 4 and the second part resembles tone 2. Native listeners, most L2 listeners in the high group, as well as the CART classification in Fig. 8 seem to be able to use the distinctive low values of coef-a to correctly identify these files as tone 3.

There are three tone 4 files with item difficulty scores below 50 for the low group, and they are circled in Fig. 15. All tone 4 files are plotted with coef-c on the x-axis and coef-b on the y-axis. Item difficulty scores for the low group are used as plotting symbols. These files have coef-b values near 0 and coef-c values near -0.005. In combination, these features describe a tone contour with a low degree of curvature and a relatively shallow slope. Though the tone shapes are still distinct from those of tone 1, L2 listeners in both the low and mid groups tend to mistake them for tone 1. There are many good examples of tone 4 with coef-c values near 0 (low curvature). If this appears in combination with a negative coef-b with a greater magnitude, then the toneshape is a straight falling line, which does not appear to cause much of a problem for L2 listeners.

In summary, this study reports robust TLD effects on Mandarin monosyllabic speech, with special attention to tone production and perception. Log intensity and duration increase systematically with increasing TLD. Tonal contours are normalized by duration and fitted with a quadratic function to obtain three tone coefficients that represent the entire  $f_0$  contour. One acoustic landmark of tone, the

maximum  $f_0$ , is also investigated. Talker-to-listener distance has a clear effect of raising maximum  $f_0$  and coef- $a$ , the coefficient that represents initial tone height. The two tone coefficients that represent tonal contours, coef- $b$  and coef- $c$ , remain invariant under changes in TLD after normalization of duration. The findings have strong implications for tone perception. The presence of invariant tone features allows native speakers to achieve excellent performance on tone recognition tasks.

Second-language learner tone perception was also investigated, and tone perception errors from the high, mid and low L2 groups were compared. Tone stimuli that are particularly difficult for L2 learners typically have tone coefficients properties that can partially explain the errors.

Tone 2 and tone 3 make up the most difficult tone pair for L2 learners and native listeners alike to identify, and speech projected to farther TLDs improves L2 tone 3 perception. It appears that the low group has not yet acquired the sensitivity to tone features that differentiates between the tones in this pair. With more advanced skills, the mid and high groups show sharpened attention to acoustic details that allow them to achieve better performance. The improvement in tone 3 identification with increasing TLD suggests that TLD-oriented speech can be effectively incorporated into L2 tone training to help learners differentiate tone 3 from tone 2, the most challenging task in Mandarin lexical tone learning.

## 8. Conclusion

We speak differently based on the distance from our listener. When the listener is farther away, we instinctively increase vocal effort, which triggers many changes in the acoustic correlates of speech, including higher intensity, longer duration and higher  $f_0$ s.

The effect of talker-to-listener distance (TLD) on  $f_0$  is of particular interest in the study of tone languages, in which  $f_0$  changes may impact lexical identity. In this paper, we conducted both production and perception studies to investigate the effects of TLD on tones and determine whether these effects will affect tone perception by native and second language listeners.

The results of the production study show that some acoustic attributes vary along with TLD and some remain invariant. This finding has strong implications for tone perception. We used quadratic decomposition to obtain three coefficients capable of representing the  $f_0$  curve. The acoustic attributes that change with TLD include intensity, duration, maximum  $f_0$  and coef- $a$ , a variable that captures the initial  $f_0$  of a tone. Two of the acoustic attributes are invariant under changes in TLD: coef- $b$  and coef- $c$ , two tone coefficients that capture the tone shape after duration normalization. The preservation of at least some information about lexical tones enables robust tone recognition by native listeners and the L2 listeners in the high performing group.

TLD-oriented speech can be used effectively to serve different purposes in language teaching, making speech stimuli easier for pronunciation training, or more difficult for testing. Mandarin speech projected to farther TLDs help L2 listeners identify tone 3, one of the most difficult tone recognition tasks. Tone 3 has two variants in natural speech, each leads to a different type of L2 error. One variant is realized with a downward pitch movement followed by a rising tail, which sounds similar to tone 2. Speech projected to farther TLDs has higher pitch ranges and enhanced contrast between high and low tone registers. Together with longer duration and higher intensity, this type of speech provides clearer cues distinguishing tone 3 from tone 2. Another tone 3 variant lacks the rising tail, which sounds similar to tone 4. This variant is rare in speech projected to farther TLDs possibly due to increased vocal effort.

This work investigates the relationship between speech production and perception when acoustic attributes change naturally in response to the demands of speech communication. The effects of TLD on speech production are systematic, and we present models and analyses with sufficient detail to simulate them. The results have applications in speech synthesis, pronunciation training and second language testing.

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